

DTCP updates

Jeremiah Dados

Integration of the project into Design 1

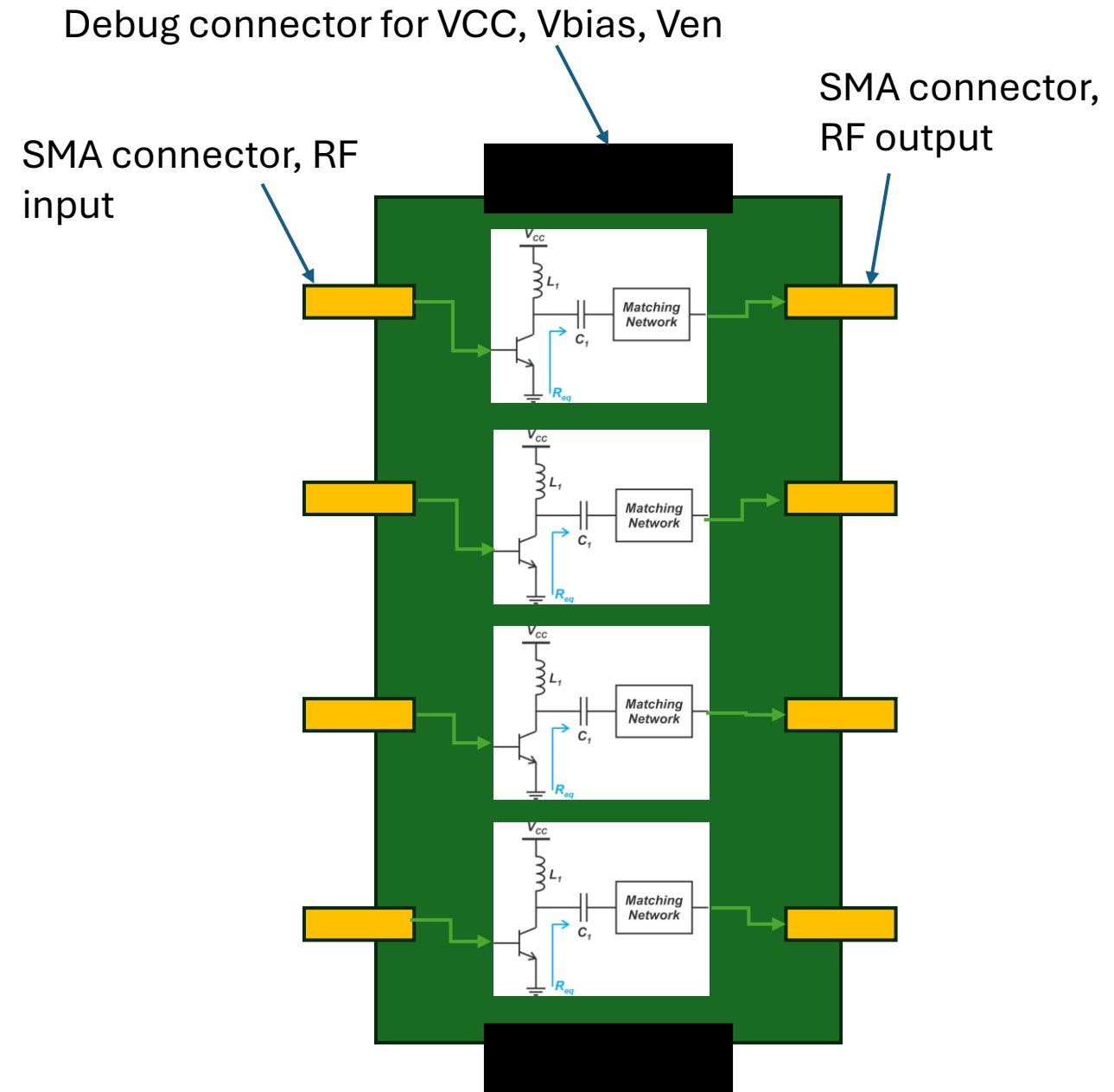
- Two new members: Luis and Dmytro
- Project timeline to meet class requirements
 - New PA circuits (up to 4W power) by mid-November
 - Basic EMG recording by mid-November
 - Combine 4W and EMG by mid-December
- PA research led by Luis, supported by Jeremiah
- EMG research led by Dmytro, supported by Jeremiah and Jay
- System integration led by Jeremiah
- Animal experiments led by Han, supported by Jeremiah

Specification	Proof-of-concept value	Target value
RF output frequency	10 MHz ~ 30 MHz	10 MHz ~ 30 MHz
RF output power	1 W	4 W
Idle power consumption	100 mW	10mW
Tx power consumption	4 W	16 W
Battery life	10 min.	24 hr.
Board dimensions	70 mm x 50 mm	40 mm x 30 mm

Table 1: Hardware Specifications

PA research

- Goal: prototype several new class-A RF Power Amplifier circuits on a single PCB
- Two frequency ranges: 10 MHz – 50 MHz, 300 MHz – 500 MHz
- Two output powers: 1W and 4W
- Stage 1: find RF MOSFETS on digikey/mouser, design circuits and simulate in Ltspice (Luis)
 - <https://www.digikey.com/en/products/filter/transistors/fets-mosfets/rf-fets-mosfets/285>
 - <https://www.mouser.com/c/semiconductors/wireless-rf-semiconductors/transistors-rf/rf-mosfet-transistors/?srsltid=AfmBOoqNT8BbQKyaANgllk4vNcVml6iil1IKQHsU4RSgrEKTZtqm00Sa>
- Stage 2: layout PCB (Jeremiah)



EMG research

- Goal: build an EMG recording system
- First using an evaluation board (Dmytro)
- Write firmware for EMG measurements (Dmytro)
- Work with Han to perform the measurements on a rat using the evaluation kit (Dmytro)
- Then integrate into PCB (Jeremiah)

ADS1299EEGFE-PDK

ADS1299 Performance Demonstration Kit

Order now

[Overview](#) | [Order & start development](#) | [Technical documentation](#) | [Support & training](#)

Overview

[Description & features](#) | [Supported products](#)

The ADS1299EEGFE-PDK is a demonstration kit for the ADS1299, an eight channel, simultaneous sampling, 24-bit, delta-sigma ($\Delta \Sigma$) analog-to-digital converter (ADC) with a built-in programmable gain amplifier (PGA), internal reference, and an on-board oscillator. The ADS1299 contains the features commonly required for electroencephalography (EEG) applications. The ADS1299EEGFE-PDK demonstration kit is designed to expedite evaluation and system development.

Features

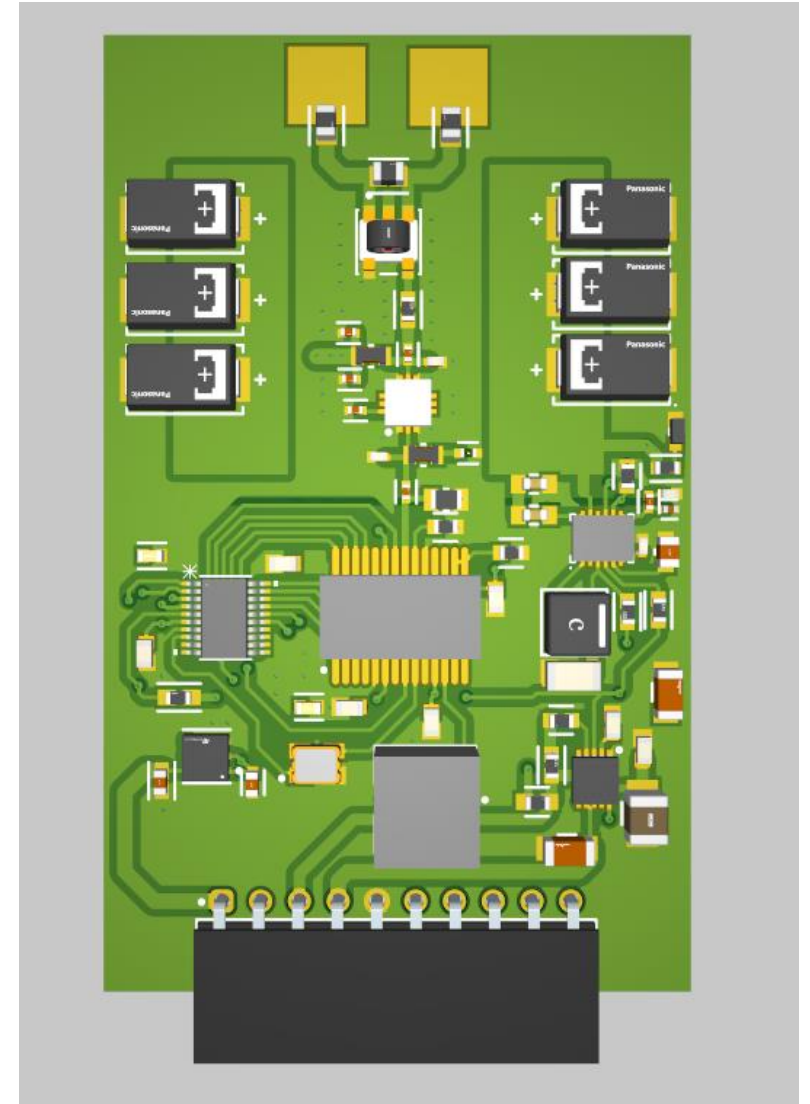
- Easy-to-use evaluation software for Microsoft™ Windows XP or Windows 7
- Built-in analysis tools including oscilloscope, FFT, and histogram displays
- Flexible input configurations and data rate options
- Ability to export data in simple test files for post processing



<https://www.ti.com/tool/ADS1299EEGFE-PDK>

Fall 2025 Final prototype

- Integrate the best PA design of the 4 created into a board similar to the first prototype
- [Link to Altium project](#) for the first prototype



DTCP updates: weeks 4-7

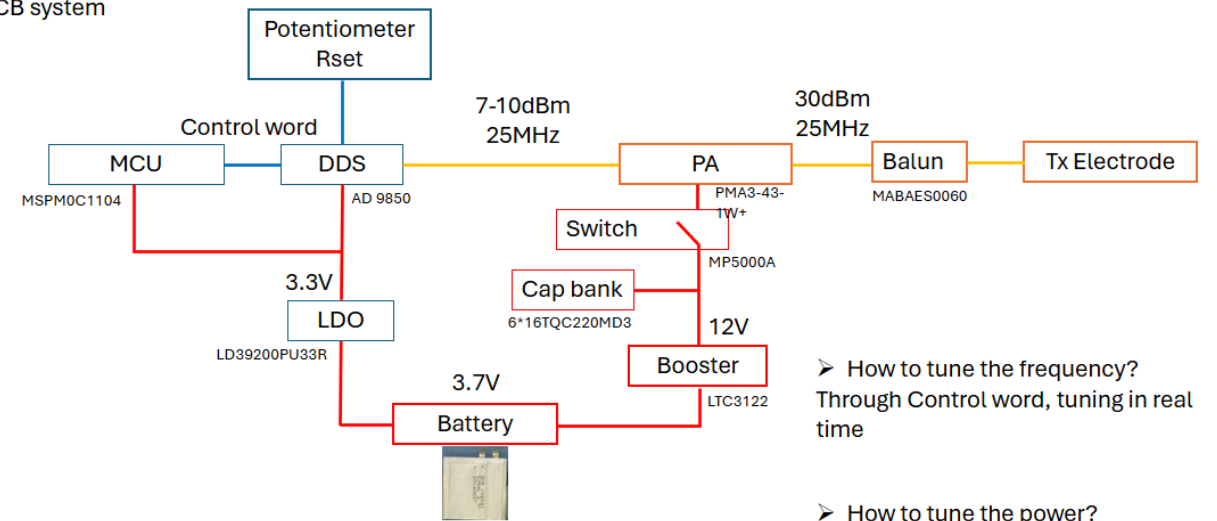
Current objectives:

- Determine if the first prototype of the board can meet the DTCP design requirements.
- If it can't, determine the necessary changes for the second prototype.



Fabricated and assembled board

Tx PCB system



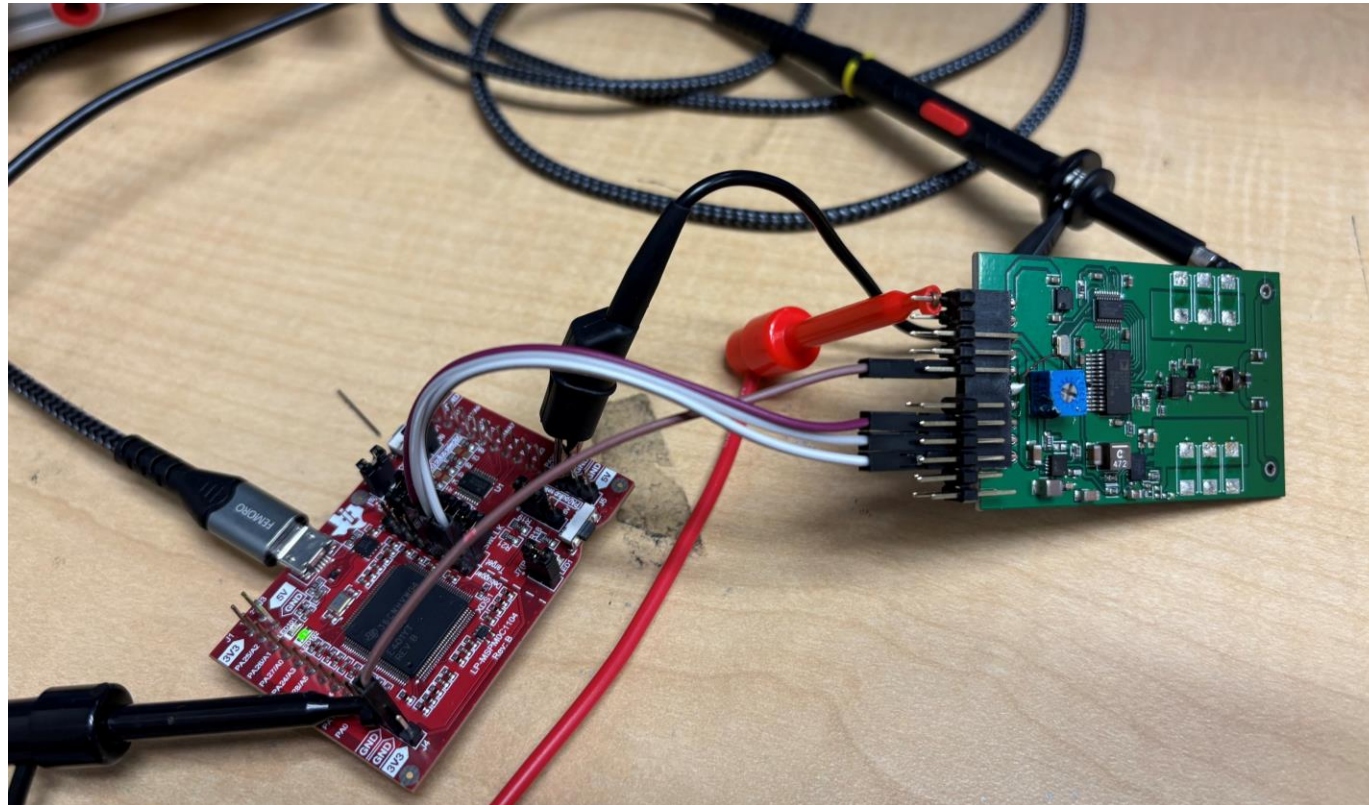
System diagram

➤ How to tune the frequency?
Through Control word, tuning in real time

➤ How to tune the power?
A potentiometer(pot) to tune Rset of the AD9850

Methods

- The board was assembled using SMD soldering on a hot plate
- Debugging was performed using the MSP1104 Launchpad and the SDS 1202 Oscilloscope.
- The board was powered using the SPD3303 DC power supply in the place of battery.



Results part 1

- The equipment currently available at the lab is not adequate for SMD PCB assembly. The hot plate is made for heating up chemical solutions, not assembling electronics. The SMD rework heat gun we've been using since last year is broken and the other heat gun isn't strong enough.
- There is no RF oscilloscope we have currently access to. The SDS1202 oscilloscope is not adequate for measurements of pulsed RF signals, because the memory depth isn't sufficient to capture them as sine waves when the time scale is increased



Broken



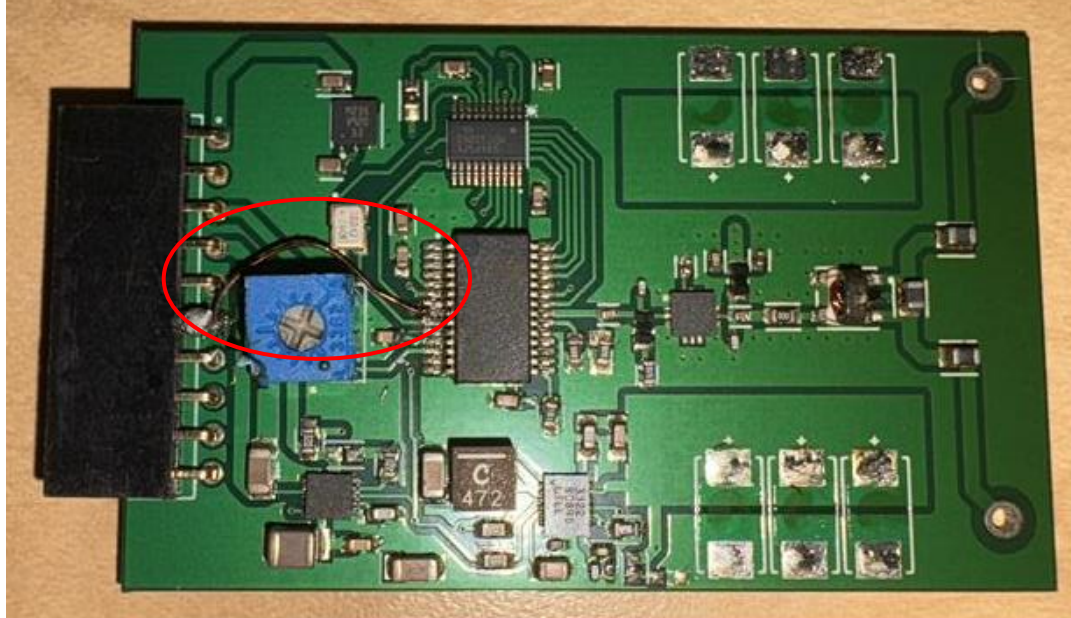
Not suitable



Not suitable

Results part 2

- It's very challenging to debug the board. Because we had to rush to make a “publication ready” board, the design doesn't have SMAs or debug connectors, and the parts are close together because we attempted to minimize board area. It takes many hours to isolate and effectively debug any part of the device.
- Assembly produces inconsistent results. The circuits exhibit many states depending on the iteration number of resoldering.
- Multiple static current consumption levels (40 mA, 80 mA, 100 mA, 120 mA, 200 mA) have been observed.
- Multiple output waveform shapes, amplitudes, and frequencies have been observed.



Assembled PCB



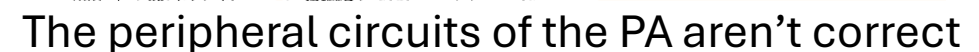
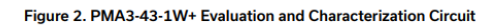
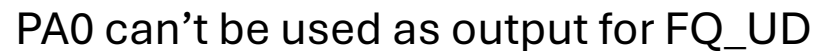
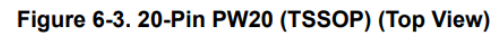
Example of output with 9Vpp 25 MHz signal

Results part 3

- Due to soldering and measurement issues, as well as due to how the board isn't designed for easy debugging, this is the status of sub-systems as of now:

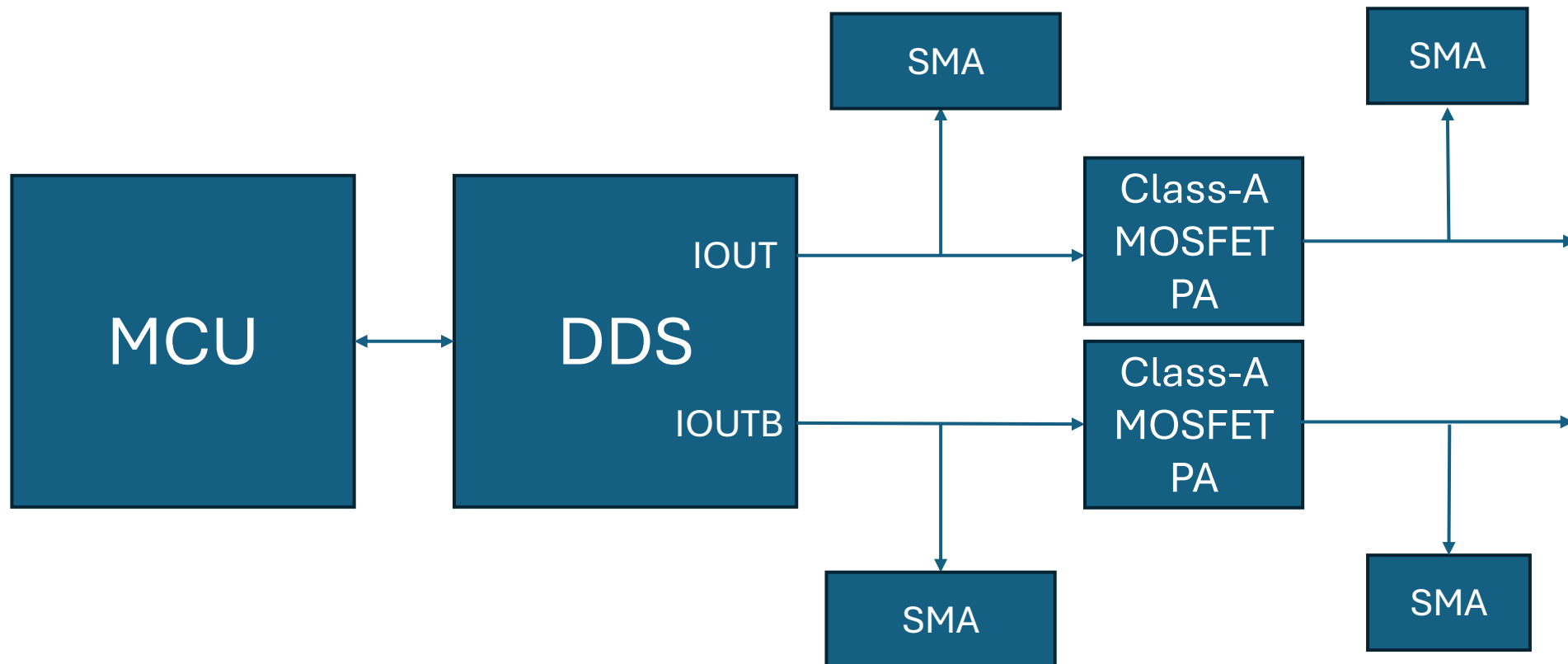
MCU	DDS	Power Switch	Boost Converter	PA	Output circuit
Fully functional	Fully functional	Functional	Functional	Unreliable	Functionality unconfirmed
After changing the pin assignment for FQ_UD, everything's working.	High static current consumption, but it can be mitigated.	The circuit fulfills the basic objective, but the efficiency hasn't been confirmed.	The circuit fulfills the basic objective, but the efficiency hasn't been confirmed.	The circuit can produce outputs after extensive modifications, but its fragile and doesn't reach full efficiency.	Due to issues with the PA, we don't know if the output circuit works.

- During debugging, I had time to more critically examine system components. Because the design verification phase of the project was rushed to meet the deadline, I wasn't able to evaluate Han's PA, power and output circuits in-depth. Overall, the following problems were detected on the board:



Results part 5

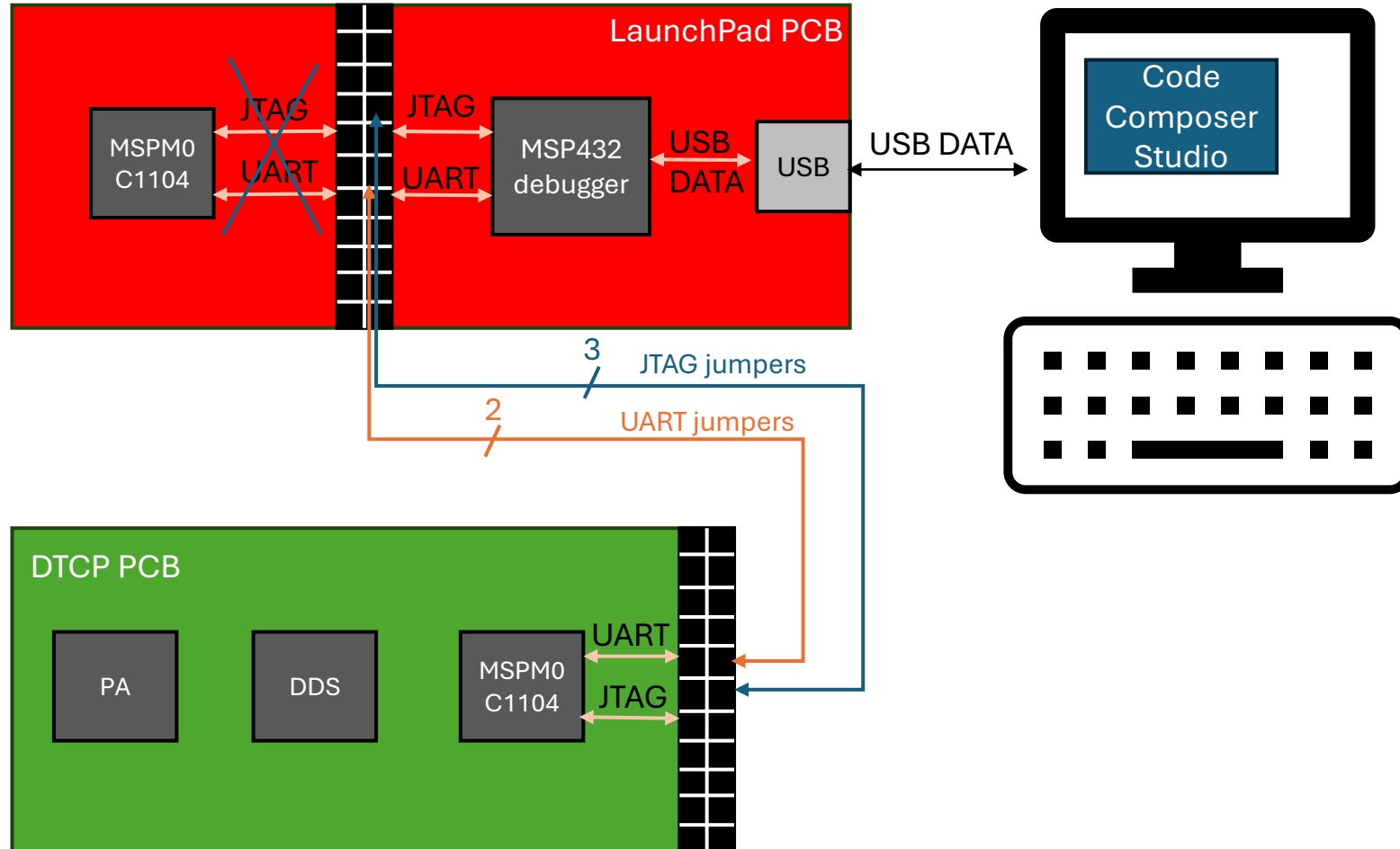
- Based on the testing and more time to think, I have reservations about the system design.
- Why aren't we using the differential outputs of the DDS, and instead we're splitting a single-ended signal into two using a balun?
- We need SMA connectors to validate the outputs. It's very inefficient to debug the system without multiple SMAs available.
- The DDS might not be the optimal solution, because it draws high standby current (>50 mA)
- I wouldn't use this PA chip. It's risky to use a newly released chip in a small QFN package. We should avoid QFN components to de-risk the designs. Also, components with reference designs available should be preferred.
- I propose the following design instead:

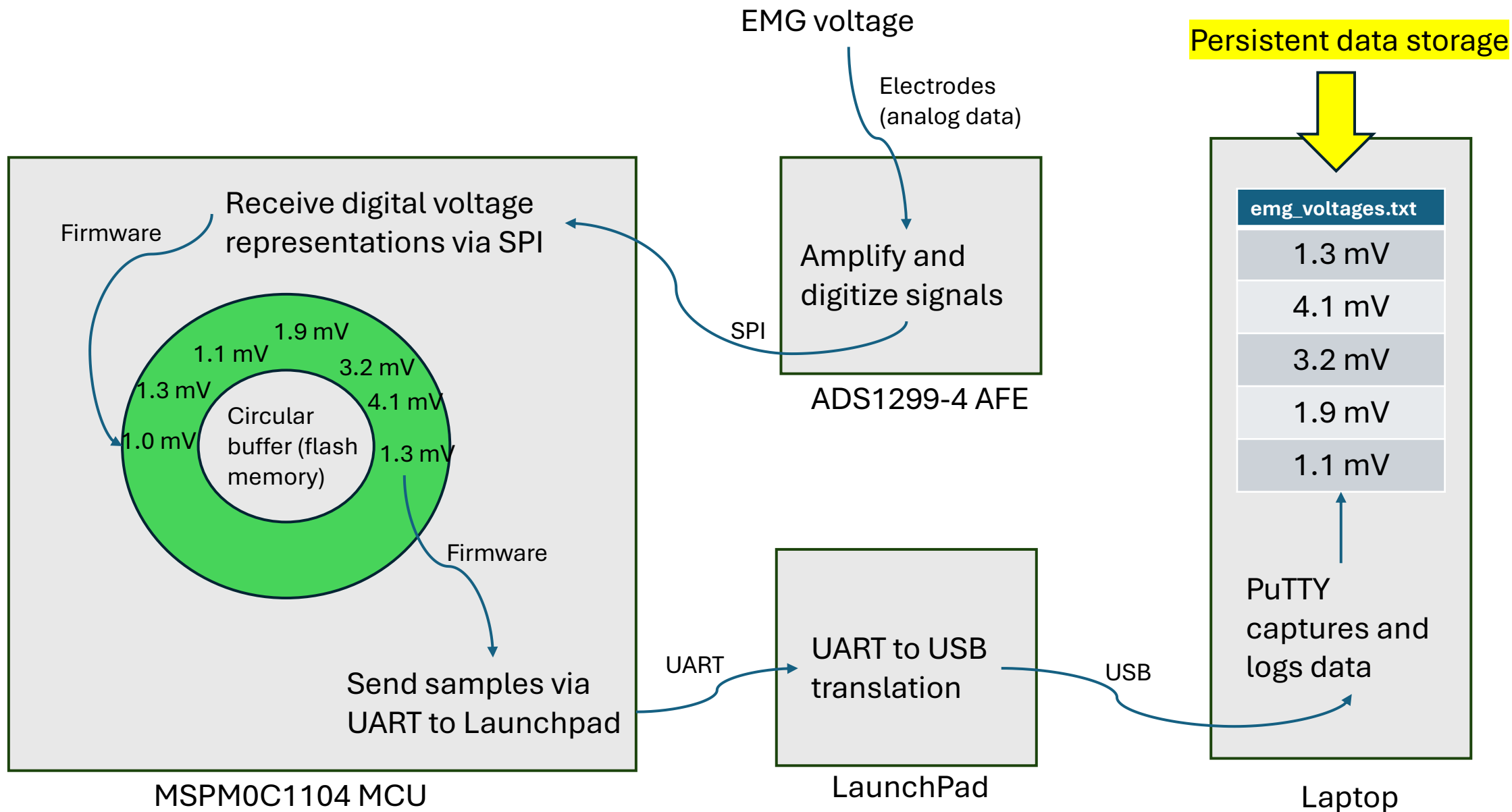


Next steps

- I don't think it's productive to keep trying to make the current board suitable for publication. It's not a good design and shouldn't be published.
- We should target another journal / conference with a deadline in January.
- In 2 weeks, we should order the following boards:
 - Tx only PCB, a fixed version of the current device
 - PA only PCB, to debug the PA and output circuits with input from signal generator
 - Tx + EMG PCB
- We need to follow engineering design principles. Iterate on designs and de-risk rather than trying to deliver the final system in one month.
- Sub-systems must be validated standalone first, and only then connected together.

Connection broken by removing the jumpers

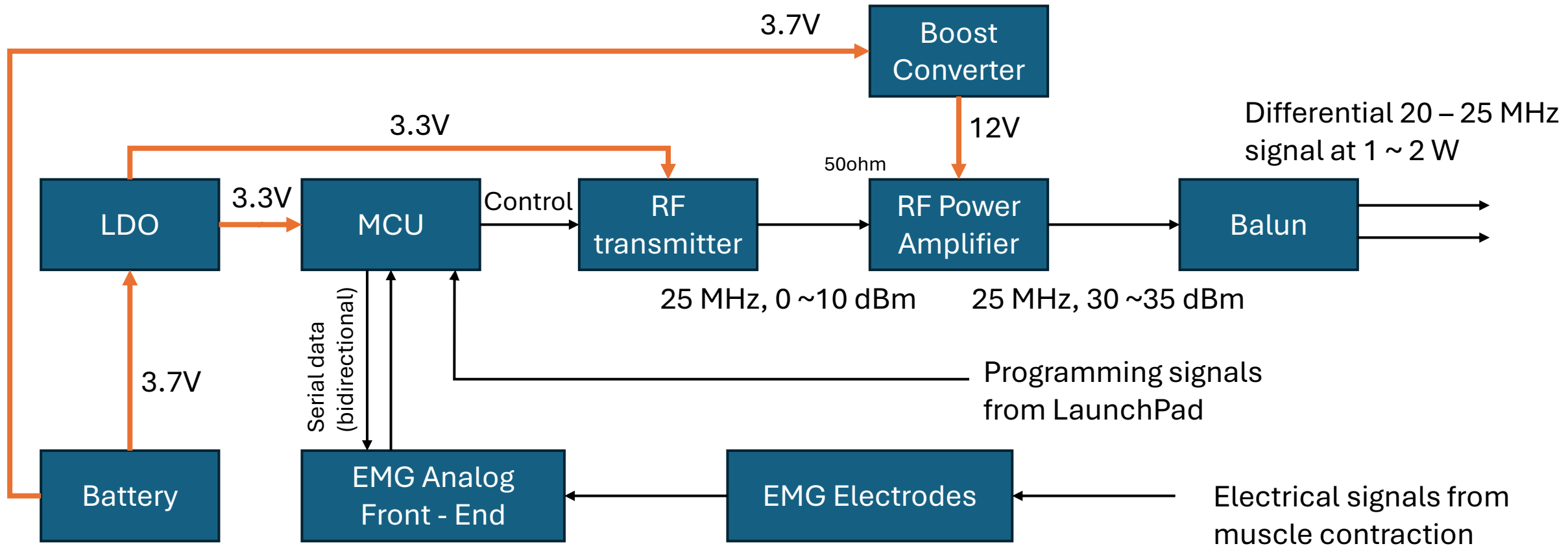




DTCP updates: week 8

Objective:

- Use the learnings from the first prototype to design the second one.



Methods

- A summary of first prototype experiments and lessons was compiled based on my and Han’s experiments.
- New systems were designed on schematic level using datasheets and Ltspice simulations (if applicable).

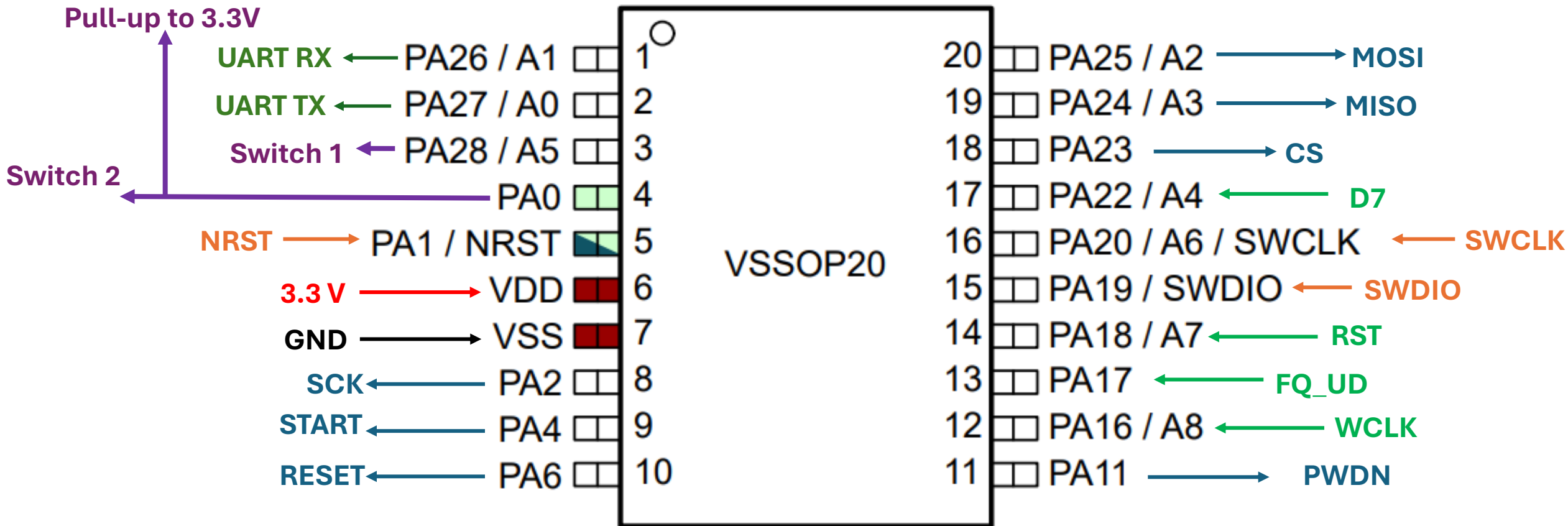
Results

- The summary of the learnings about sub-system that caused problems in the first prototype is presented below:

MCU	RF Transmitter (DDS)	Power Amplifier
• The PA1 pin cannot be used to drive any signals without a pull-up resistor • The number of GPIO pins is not sufficient to support both the DDS and EMG interfaces under the current wiring	• The standby current is too large to allow for battery operation (80 mA) • The differential output capabilities aren’t used • The output power is less than expected (6 – 8 dBm)	• The matching network is wrong (not matched for 25 MHz) • The footprint is wrong (no GND connections for pins) • The operation is unreliable • In the rare cases that the circuit works, the output power saturates at 28 dBm

New MCU wiring

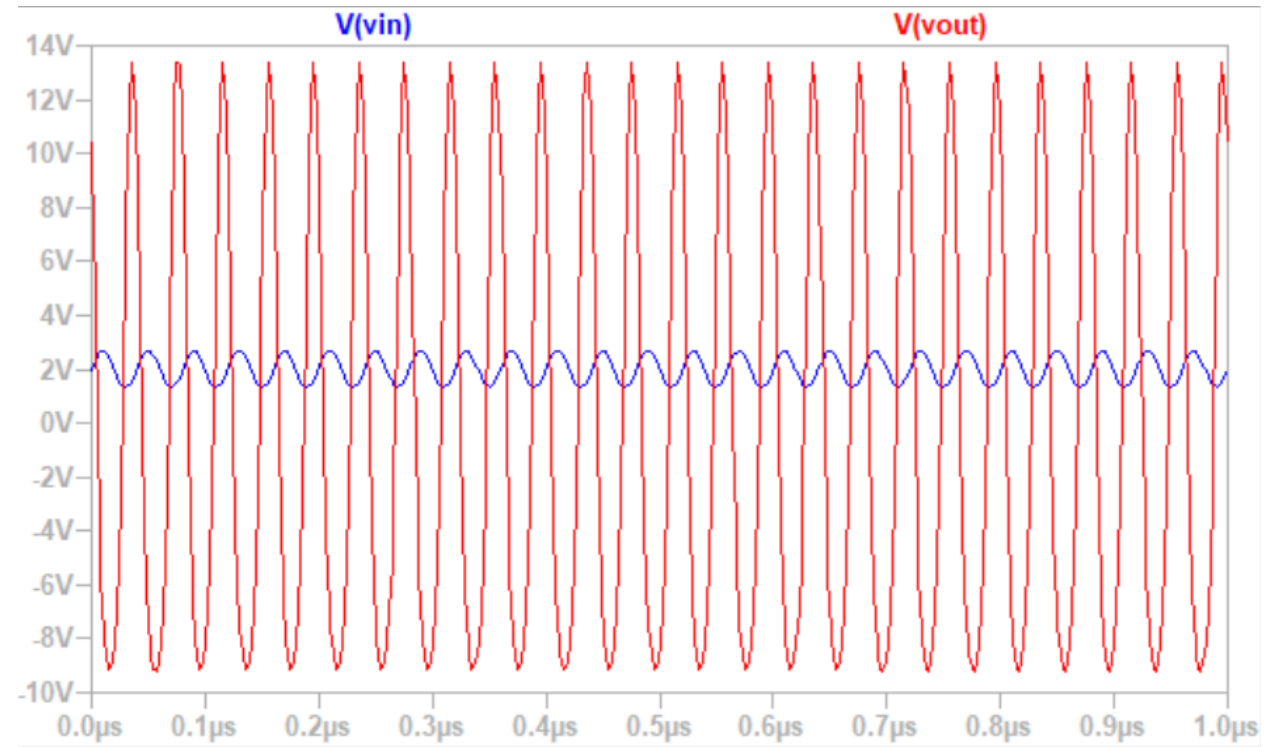
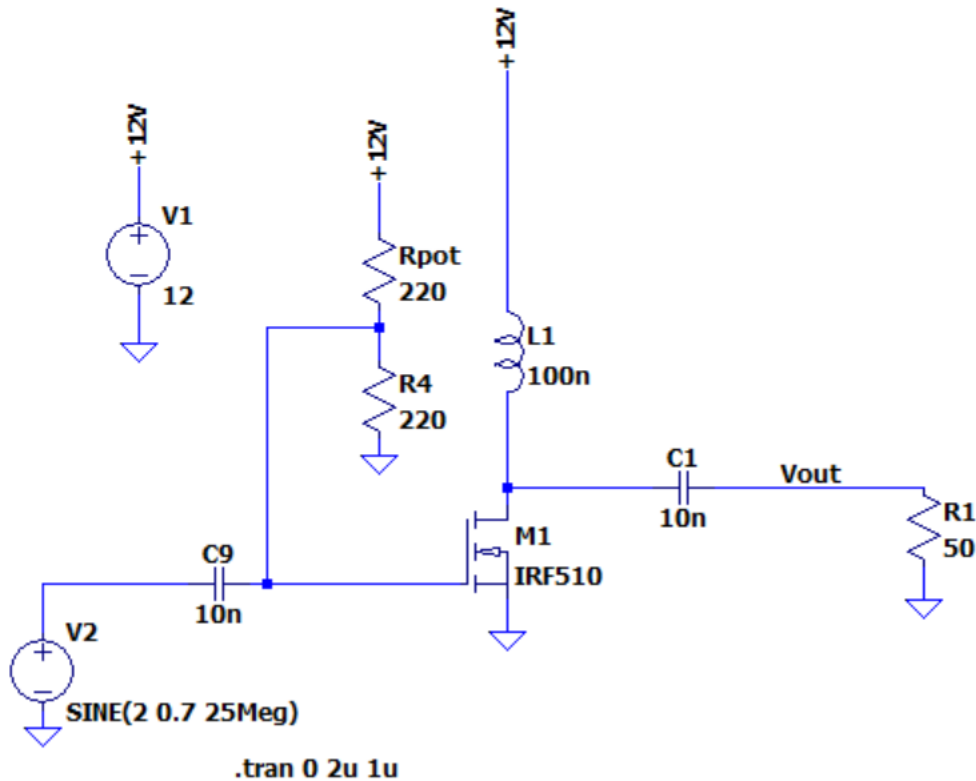
- At least 7 pins needed to program the **DDS**
- At least 7 pins needed to program the **EMG AFE**
- One GPIO needed to control power to other components (assuming we switch all PAs and DDS on and off at the same time).
- At least 3 pins needed for **JTAG** and 2 for power and ground



Note: with this wiring, we can no longer use UART to debug the system. Do we need UART to print EMG data, or is it enough to just store data in memory like with the EEG device?

New Power Amplifier Solutions

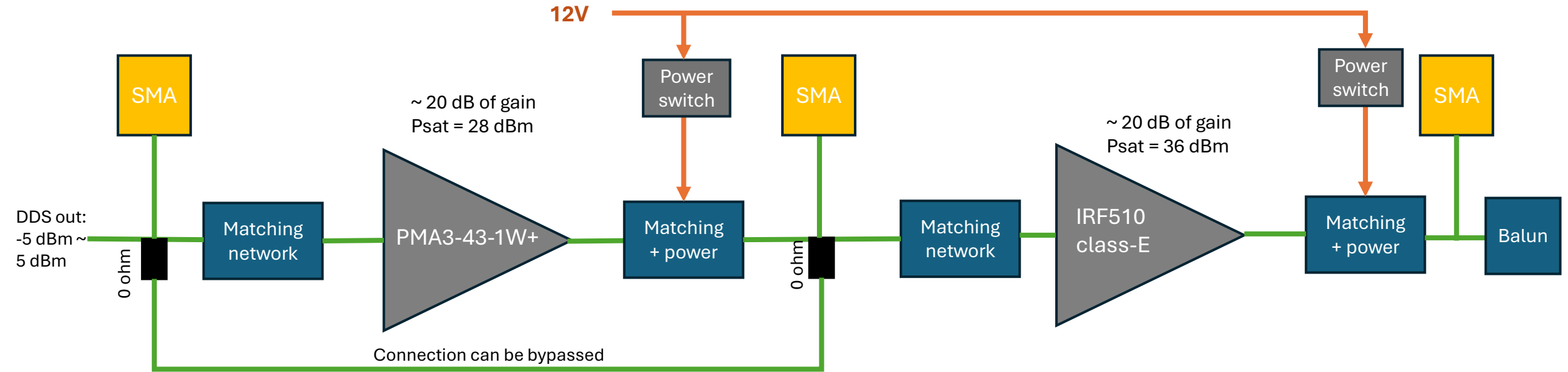
- The previously evaluated PA was unreliable and only delivered 28 dBm of power.
- However, it's the only commercially available PA in our frequency range that saturates at ~1W.
- Because of that, we need to use a custom class-E MOSFET-based PA for the last stage.
- IRF510 is a promising candidate based on a TI reference design and Luis is researching the details of the implementation we need.



IRF510 in a basic class-E configuration amplifies a 7 dBm (1.4 Vpp) signal to 31 dBm (22 Vpp)

New Power Amplifier Architecture: version 1

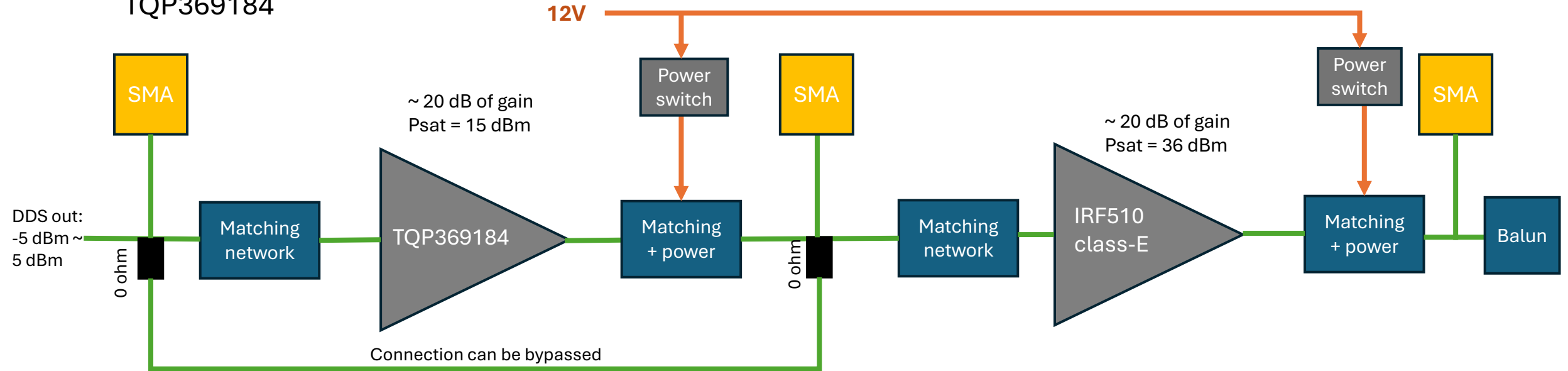
- The following topologies are proposed to deliver a low-risk, easily debuggable design.
- Two versions of the board will be fabricated to minimize risk and ensure at least one design produces the desired output.



Generic photo used for illustration purposes only

New Power Amplifier Architecture: version 2

- PMA3-43-1W+ has two issues: It uses a tiny QFN package, making it hard to assemble and debug, and consumes a large quiescent current during standby.
- I'd like to use another, simpler PA because the first stage doesn't need a 28 dBm input. For example TQP369184



SOT-363 Package

Product Features

- DC – 6000 MHz
- Flat, broadband frequency response
- 20.3 dB Gain at 1900 MHz
- 3.9 dB Noise Figure at 1900 MHz
- +28.5 dBm Output IP3 at 1900 MHz
- +15.5 dBm P1dB at 1900 MHz
- 50 Ohm Cascadable Gain Block
- Single Supply, 45 mA Current
- SOT-363 Package

DTCP week 9 – Jeremiah Dados

Objective:

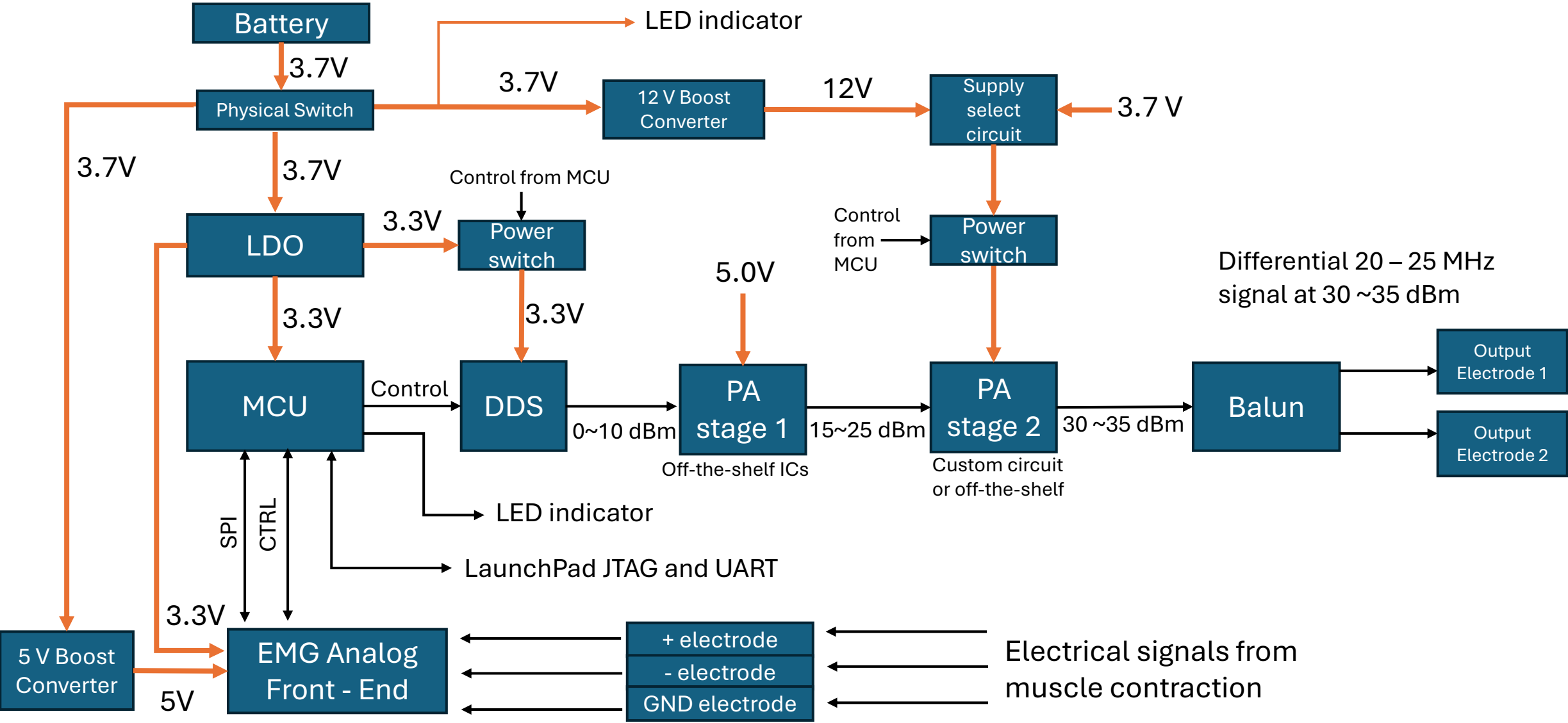
Design the second prototype of the DTCP transmitter board (fix errors, new PA design, EMG recording added)

Methods:

- Digikey and Mouser component search was utilized to find integrated circuits capable of realizing the system architecture within the design constraints. Datasheets were carefully examined for each part to identify application circuits and required external parts.
- LTspice was used to test the PA design
- Altium Designer was used for preliminary schematic design of the PCB

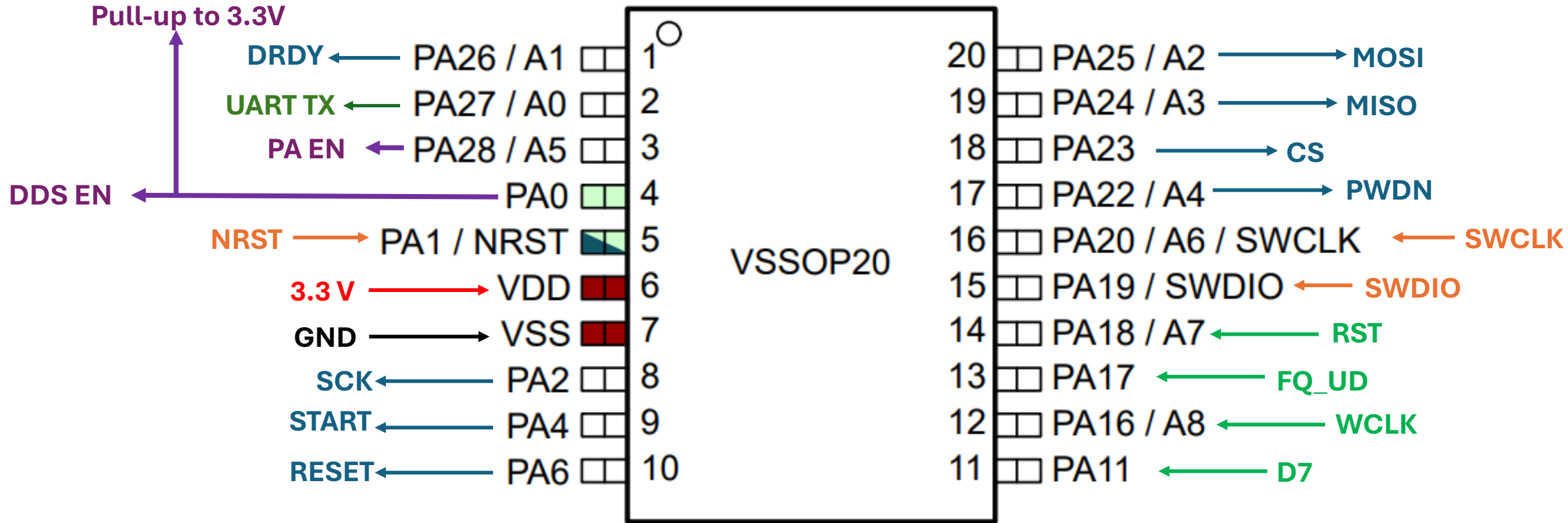
Results

Updated system architecture diagram:

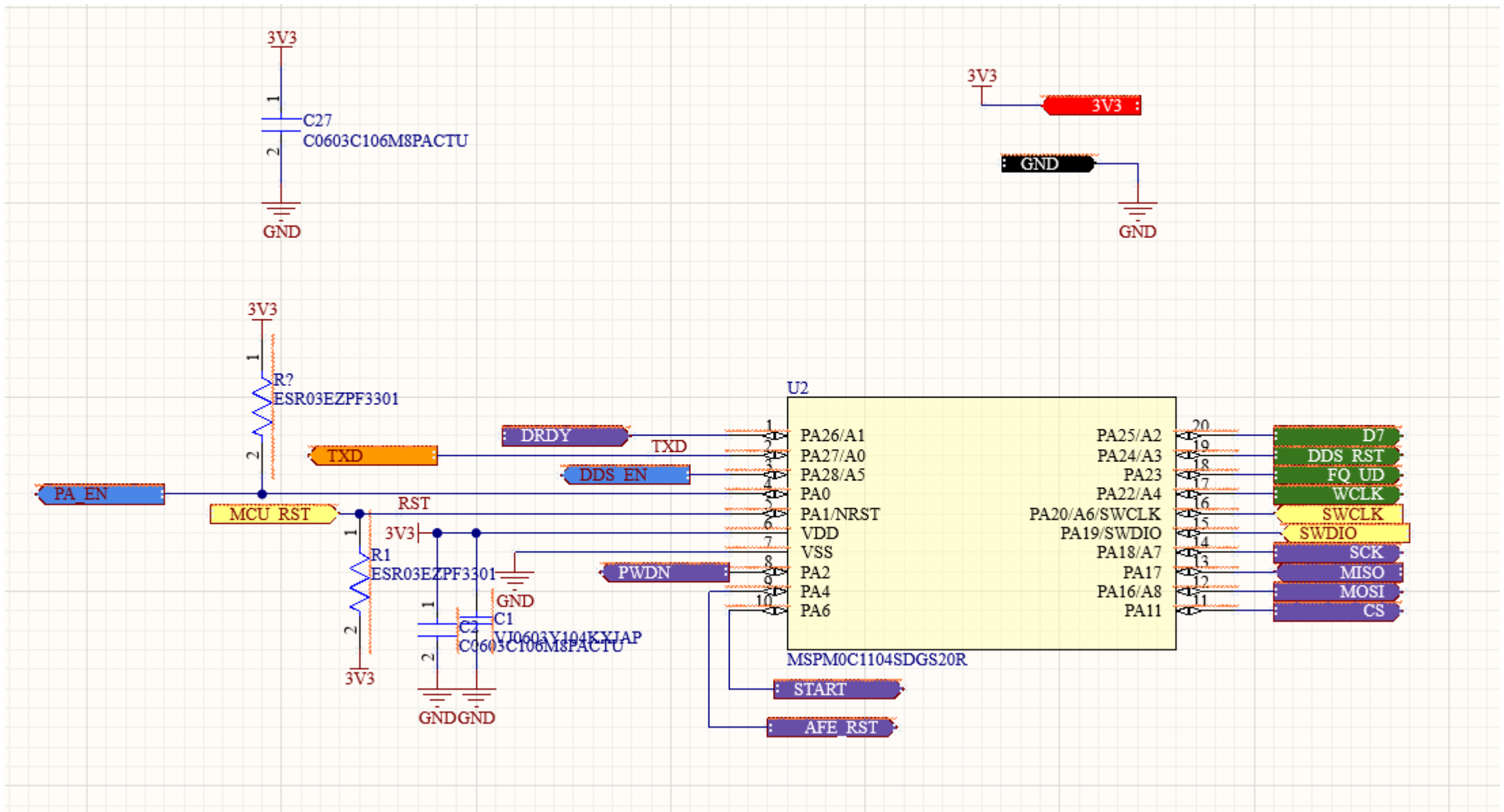


MCU changes

- There is one AFE pin we didn't account for: DRDY. To use it, we need to sacrifice one GPIO on the MCU.
- We don't actually need UART RX, since the laptop won't send data to the board



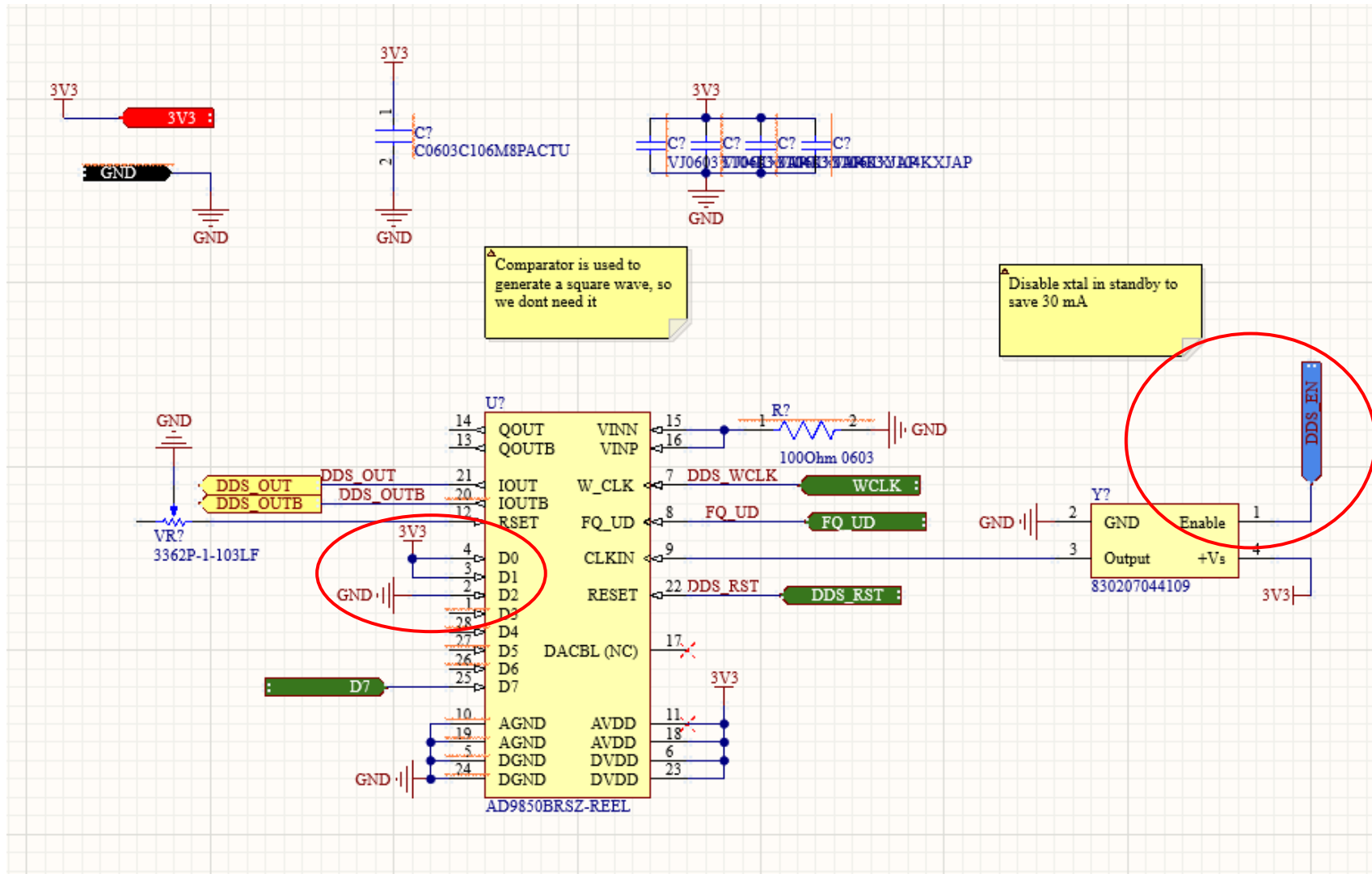
New wiring diagram of the MCU



New Altium schematic of the MCU circuit

DDS changes

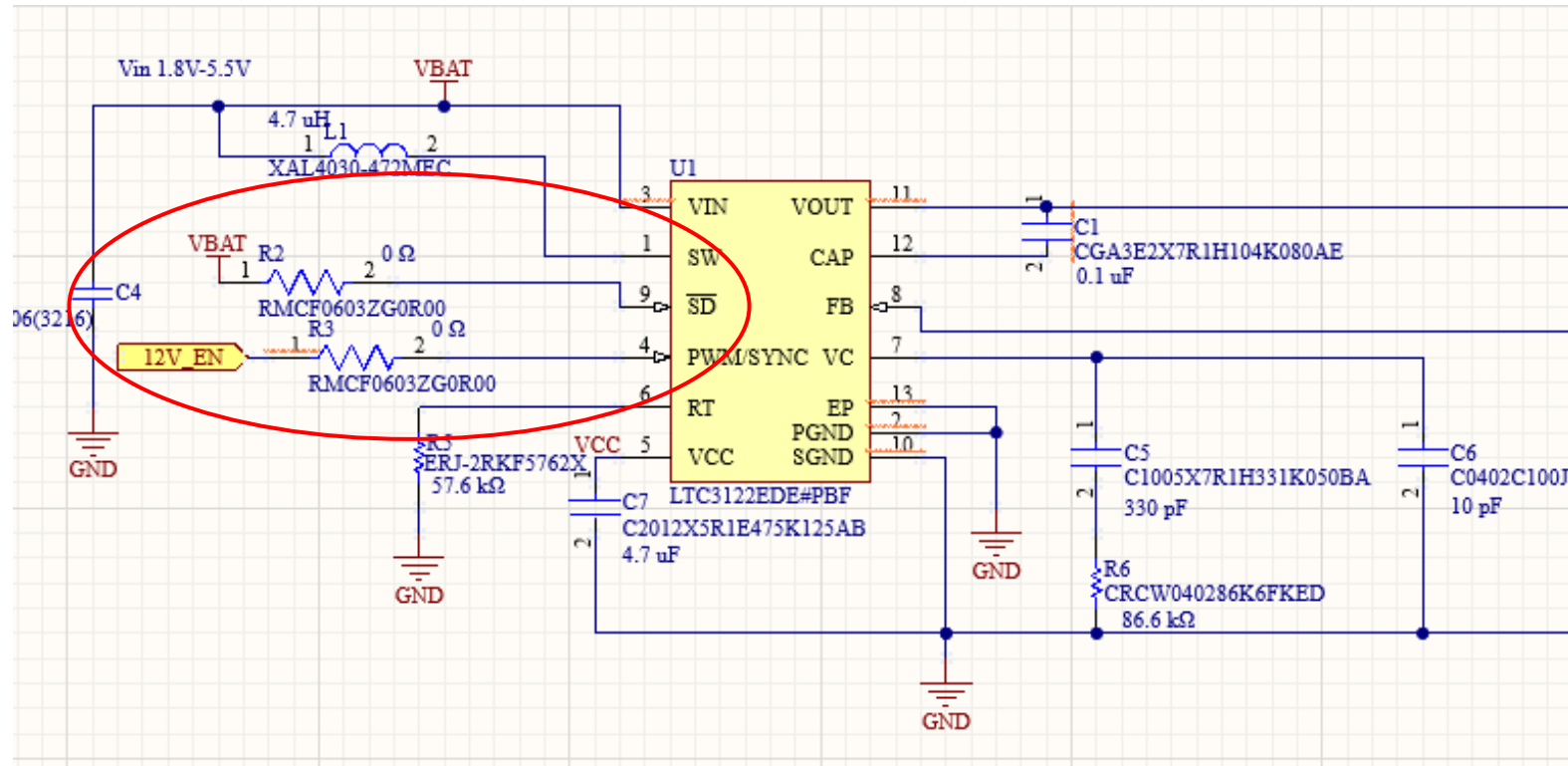
- We will hardwire D0-D2 to 011 to save GPIO pins on the MCU.
- I added a previously missing EN connection to the 100 MHz oscillator to save 30 mA of static current.



New Altium schematic of the DDS circuit with changes highlighted

Boost converter changes

- The schematic of the 12V boost converter wasn't correct before. The SW and SD pins weren't connected as shown in the application circuit. I fixed that in this version.
- The same IC will also be used to generate 5 V for the 1st stage PA and the EMG AFE



Previous boost converter circuit

PWM/SYNC (Pin 4): Burst Mode Operation Select and Oscillator Synchronization. **Do not leave this pin floating.**

- PWM/SYNC = High. Disable Burst Mode Operation and maintain low noise, constant frequency operation.
- PWM/SYNC = Low. The converter operates in Burst Mode operation, independent of load current.
- PWM/SYNC = External CLK. The internal oscillator is synchronized to the external CLK signal. Burst Mode operation is disabled. A clock pulse width between 100ns and 2µs is required to synchronize the oscillator. An external resistor **must be** connected between RT and GND to program the oscillator slightly below the desired synchronization frequency.

In non-synchronized applications, repeated clocking of the PWM/SYNC pin to affect an operating mode change is supported with these restrictions:

- Boost Mode ($V_{OUT} > V_{IN}$): $I_{OUT} < 500\mu A$: $f_{PWM/SYNC} \leq 100Hz$, $I_{OUT} \geq 500\mu A$: $f_{PWM/SYNC} \leq 5kHz$
- Buck Mode ($V_{OUT} < V_{IN}$): $I_{OUT} < 5mA$: $f_{PWM/SYNC} \leq 5Hz$, $I_{OUT} \geq 5mA$: $f_{PWM/SYNC} \leq 5kHz$

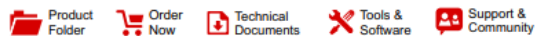
SD (Pin 9): Logic Controlled Shutdown Input. Bringing this pin above 1.6V enables normal, free-running operation, forcing this pin below 0.25V shuts the LTC3122 down, with quiescent current below 1µA. **Do not leave this pin floating.**

-

Altium schematic of the new 12 V boost converter circuit

Power switch changes

- Different IC was used to avoid a small QFN package and simplify the circuit.
- Two power switches will be present in the design: one for the DDS and one for the 2nd stage of the PA.



TPS22810-Q1
SLVSEJ0 – APRIL 2018

TPS22810-Q1, 2.7-18-V, 79-mΩ On-Resistance Load Switch With Thermal Protection

1 Features

- Qualified for Automotive Applications
- AEC-Q100 Qualified With the Following Results:
 - Device Temperature Grade 2: –40°C to +105°C Ambient Operating Temperature
 - Device HBM ESD Classification Level 2
 - Device CDM ESD Classification Level C5
- Integrated Single Channel Load Switch
- 2-A Maximum Continuous Current
- Input Voltage: 2.7 V to 18 V
- Absolute Maximum Input Voltage: 20 V
- On-Resistance (R_{ON})
 - $R_{ON} = 79\text{ m}\Omega$ (Typical) at $V_{IN} = 12\text{ V}$
- Quiescent Current
 - $62\text{ }\mu\text{A}$ (Typical) at $V_{IN} = 12\text{ V}$
- Shutdown Current
 - 500 nA (Typical) at $V_{IN} = 12\text{ V}$
- Thermal Shutdown
- Undervoltage Lock-Out (UVLO)
- Adjustable Quick Output Discharge (QOD)
- Configurable Rise Time With CT Pin
- SOT23-6 Package
 - $2.9\text{-mm} \times 2.8\text{-mm}$, 0.95-mm Pitch, 1.45-mm Height (DBV)

2 Applications

- Automotive Head Unit
- Surround View ECU

3 Description

The TPS22810-Q1 is a one channel load switch with configurable rise time and integrated quick output discharge (QOD). The device features thermal shutdown to protect the device against high junction temperature and thereby ensure safe operating area of the device inherently. The device features a N-channel MOSFET that can operate over an input voltage range of 2.7 V to 18 V. The device can support a maximum current of 2 A. The switch is controlled by an on and off input that can interface directly with low-voltage control signals.

The configurable rise time of the device greatly reduces inrush current caused by large bulk load capacitances, thereby reducing or eliminating power supply droop. Undervoltage lock-out is used to turn off the device if the V_{IN} voltage drops below a threshold value, ensuring that the downstream circuitry is not damaged by being supplied by a voltage lower than intended. The configurable QOD pin controls the fall time of the device to allow design flexibility for power down.

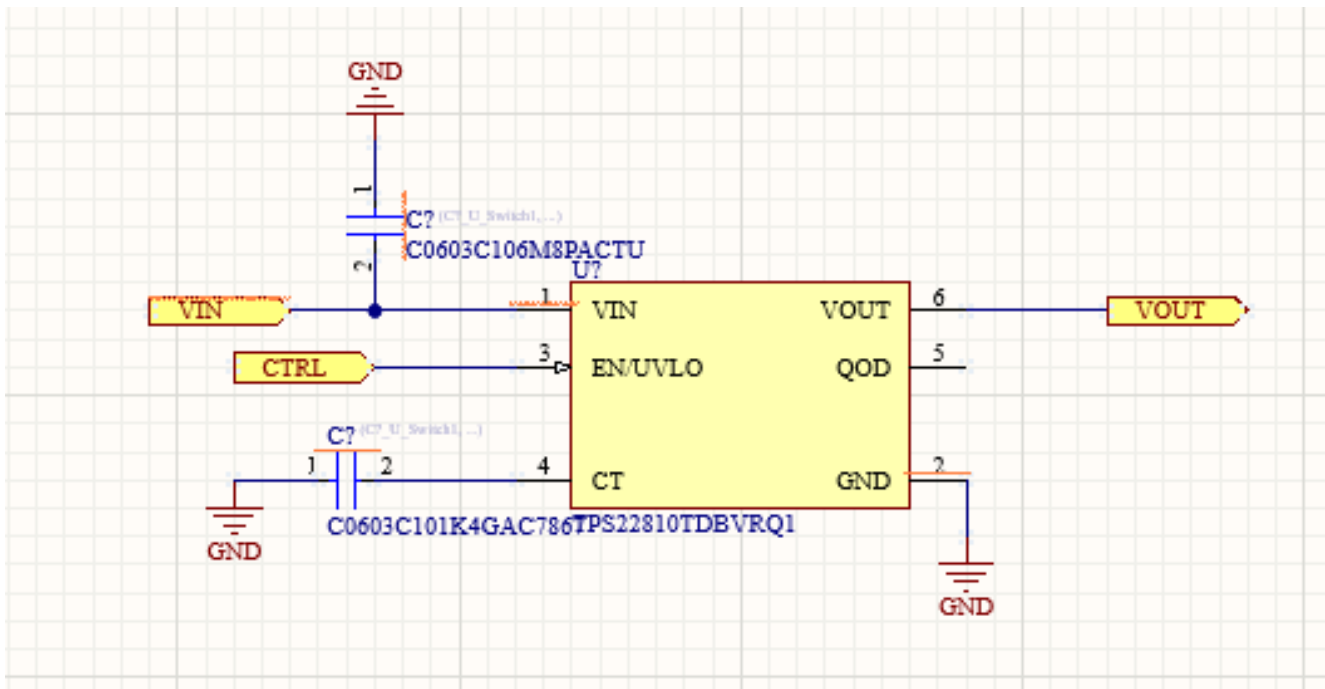
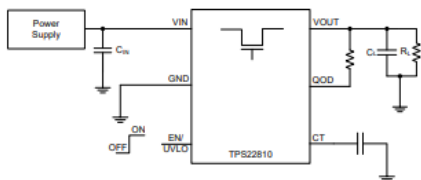
The TPS22810-Q1 is available in a leaded, SOT-23 package (DBV) which allows to visually inspect solder joints. The device is characterized for operation over the free-air temperature range of –40°C to +105°C.

Device Information⁽¹⁾

PART NUMBER	PACKAGE	BODY SIZE (NOM)
TPS22810-Q1	SOT-23 (6)	2.90 mm × 2.80 mm

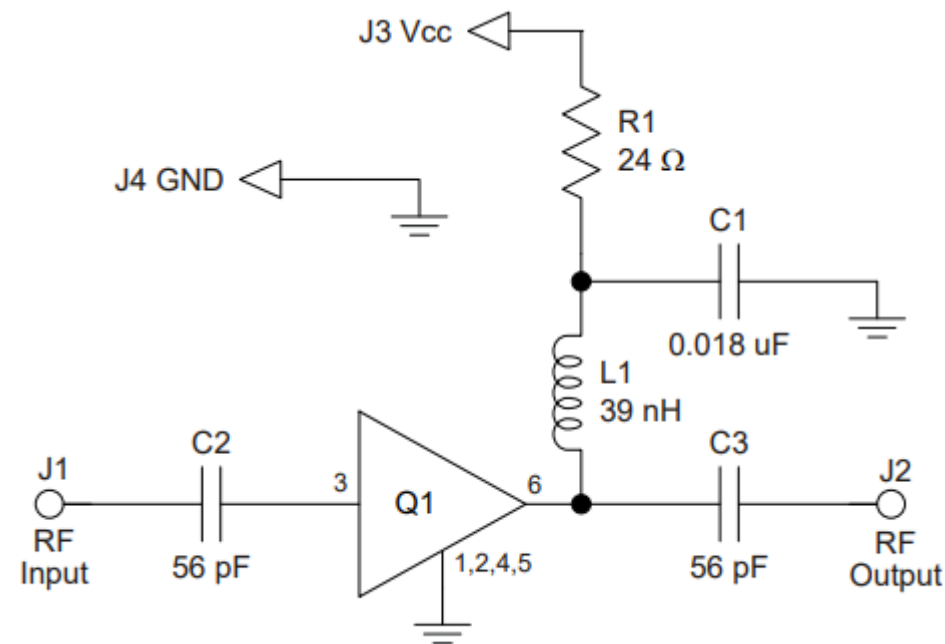
(1) For all available packages, see the orderable addendum at the end of the data sheet.

Simplified Schematic



PA: stage 1 version 1 design

- The first stage was designed using TQP369184. I chose it due to a simple application circuit and low cost.
- It will generate a ~15 dBm signal from the DDS ~5 dBm signal even with suboptimal performance



SOT-363 Package

Electrical Specifications

Parameter	Conditions ⁽¹⁾	Min	Typ	Max	Units
Operational Frequency Range		DC		6000	MHz
Test Frequency			1900		MHz
Gain		18.8	20.3	21.8	dB
Input Return Loss			15		dB
Output Return Loss			24		dB
Output P1dB			+15.5		dBm

Bill of Material – TQP369184-PCB

Reference Des.	Value	Description	Manuf.	Part Number
Q1	n/a	High Linearity LNA Gain Block	Qorvo	TQP369184
C1	0.018 uF	Cap, Chip, 0603, 16V, X7R, 10%	Various	
C2, C3	56 pF	Cap, Chip, 0603, 50V, NPO, 5%	Various	
L1	39 nH	Inductor, 0603, 5%, CS Series	Coilcraft	0603CS-39NXJL
R1	24 Ω	Res, Chip, 0805, 1/10W, 5%	Various	

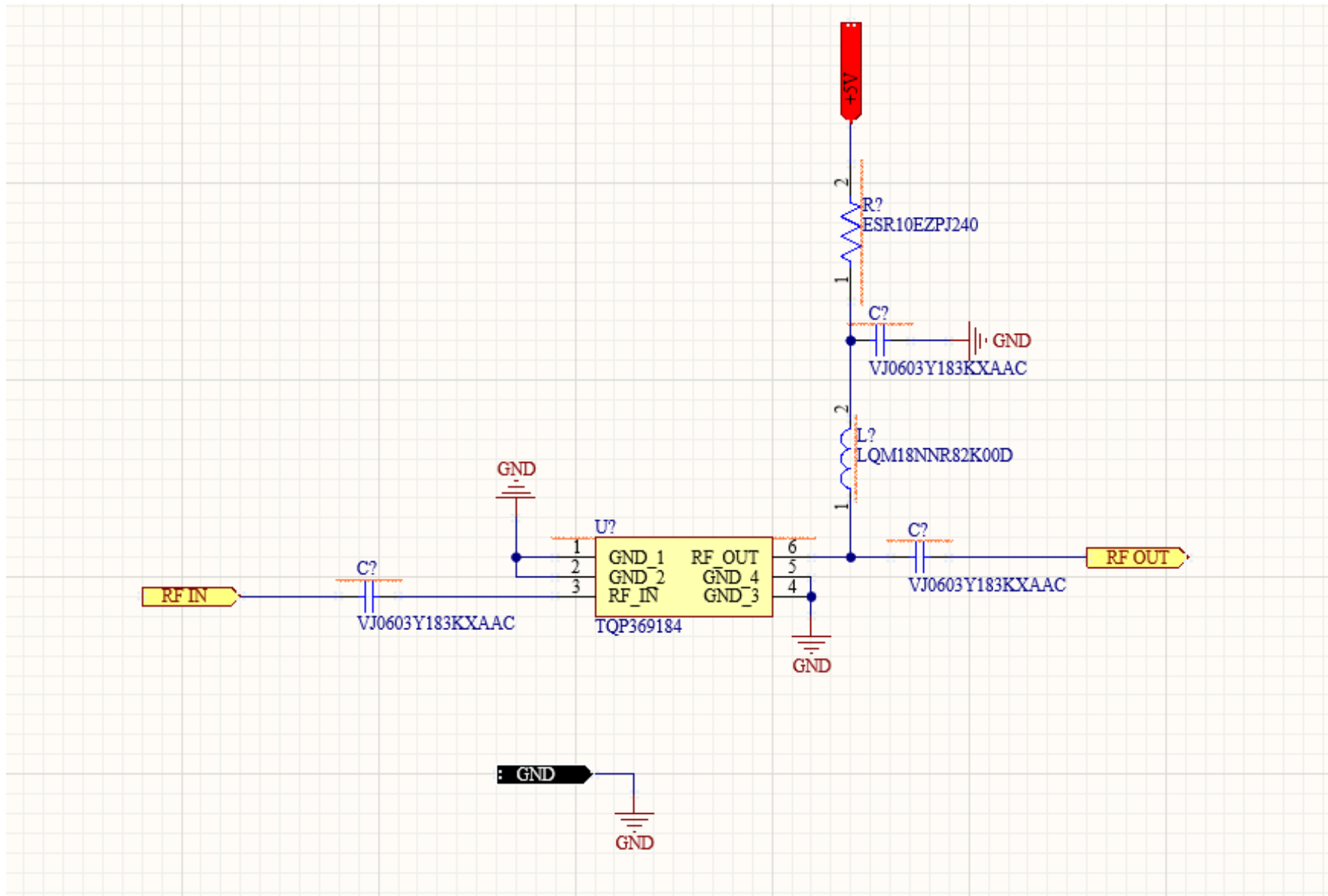
Component Values for Specific Frequencies

Use the component values in this table for optimal operation at specific frequencies.

Reference Designator	Frequency (MHz)						
	50	500	900	1900	2200	2500	3500
L1	820 nH	220 nH	68 nH	27 nH	22 nH	18 nH	15 nH
C2, C3	.018 uF	1000 pF	100 pF	68 pF	68 pF	56 pF	39 pF

Bias Resistor Values for Various Supply Voltages

V _{SUPPLY} (V)	5	6	7	8	9	10	12
R1	24 Ω	47 Ω	68 Ω	91 Ω	110 Ω	130 Ω	180 Ω
Component Size	0805	1206	1210	1210	1210	2010	2010



Altium schematic of the new 1st stage PA

PA: stage 2 version 1 design

- I'd like to use a Class-A design combined with a power switch for simplicity.
- I'm currently working on part selection. It's challenging to find RF inductors rated for high current.

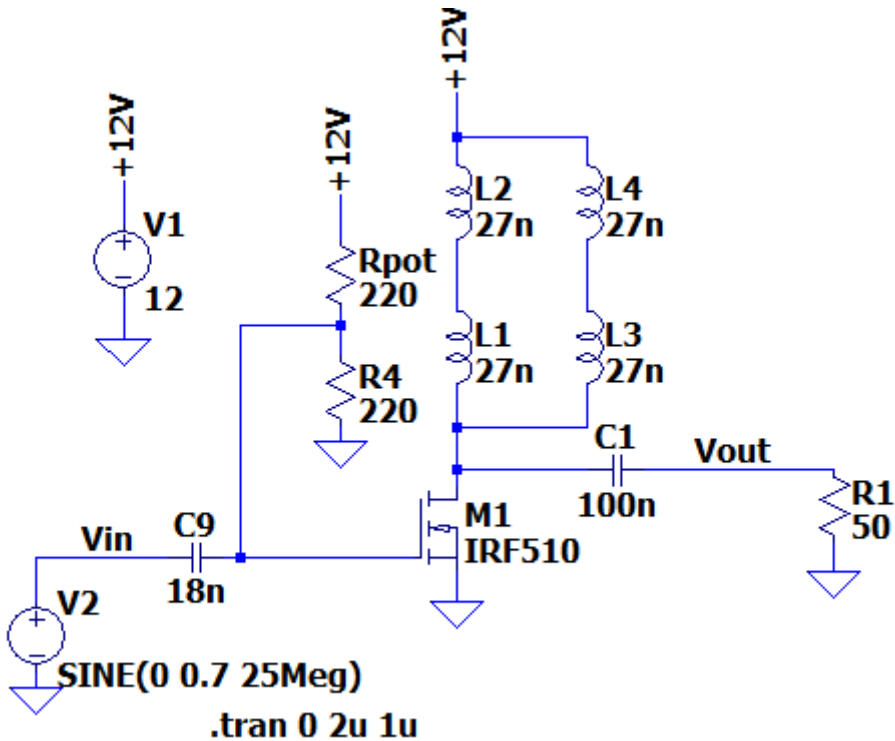
LQW2BAN6N8J00#

"#" indicates a package specification code.

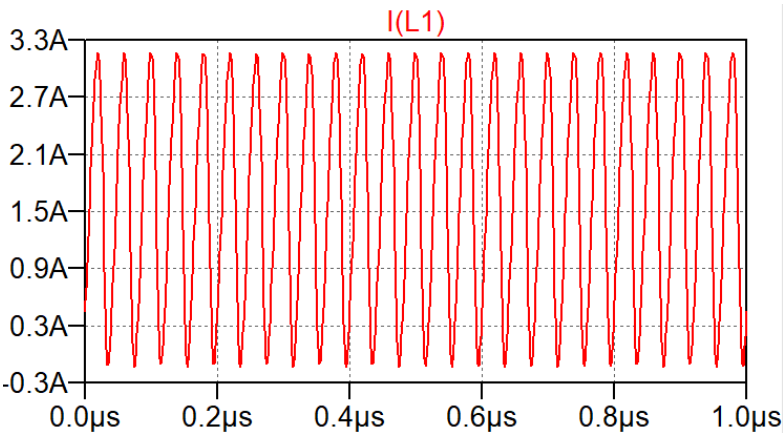
Specifications

L size	2.09±0.2mm
W size	1.53±0.2mm
T size	1.42±0.1mm
Size code inch (mm)	0805 (2015)
Inductance	6.8nH±5%
Inductance Test Frequency	250MHz
Rated current (Itemp) (Based on Temperature rise)	3000mA
Max. of DC resistance	0.03Ω
Operating Temperature Range (Self-temperature rise is included)	-55°C to 125°C
Q(min.)	90
Q Test Frequency	1000MHz
Self resonance frequency (min.)	6200MHz
Series	LQW2BAN_00

The best inductor I found is rated for 3A



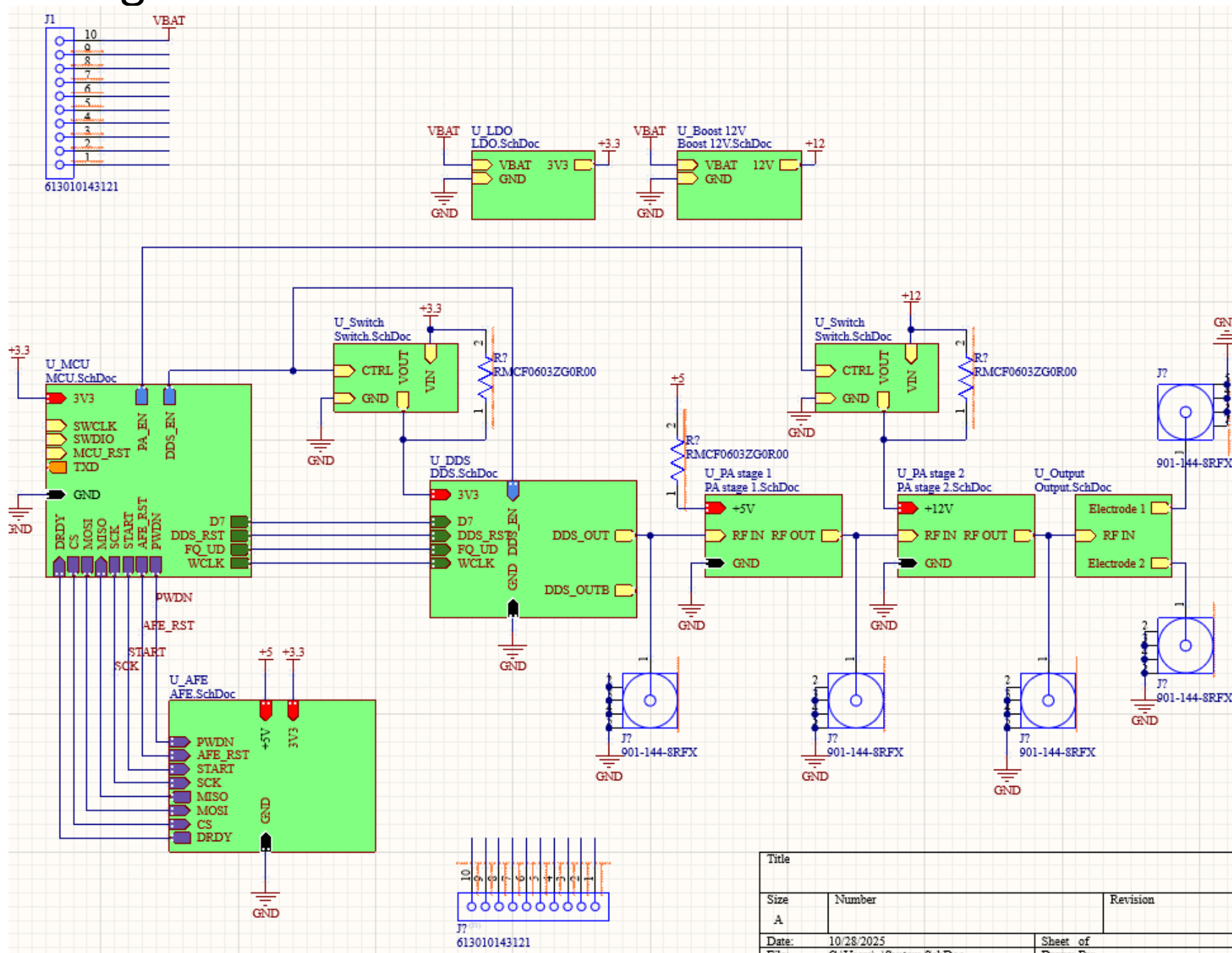
Multiple inductors in parallel might be needed



In the case of 2 in parallel, the current still exceeds 3A per inductor

High-level PCB schematic design

- In this version, I'd like to prioritize debug and ease of testing over small size.
- SMA connectors will be added at each RF output.
- Headers will be used to have debug access to all digital signals.
- Switches can be bypassed with 0 ohm resistors
- This way, I can guarantee with 90% confidence this design will work as opposed to 10% confidence I had with the last board.



DTCP week 10 – Jeremiah Dados

Objective:

Finalize the design of the DTCP prototype.

Methods:

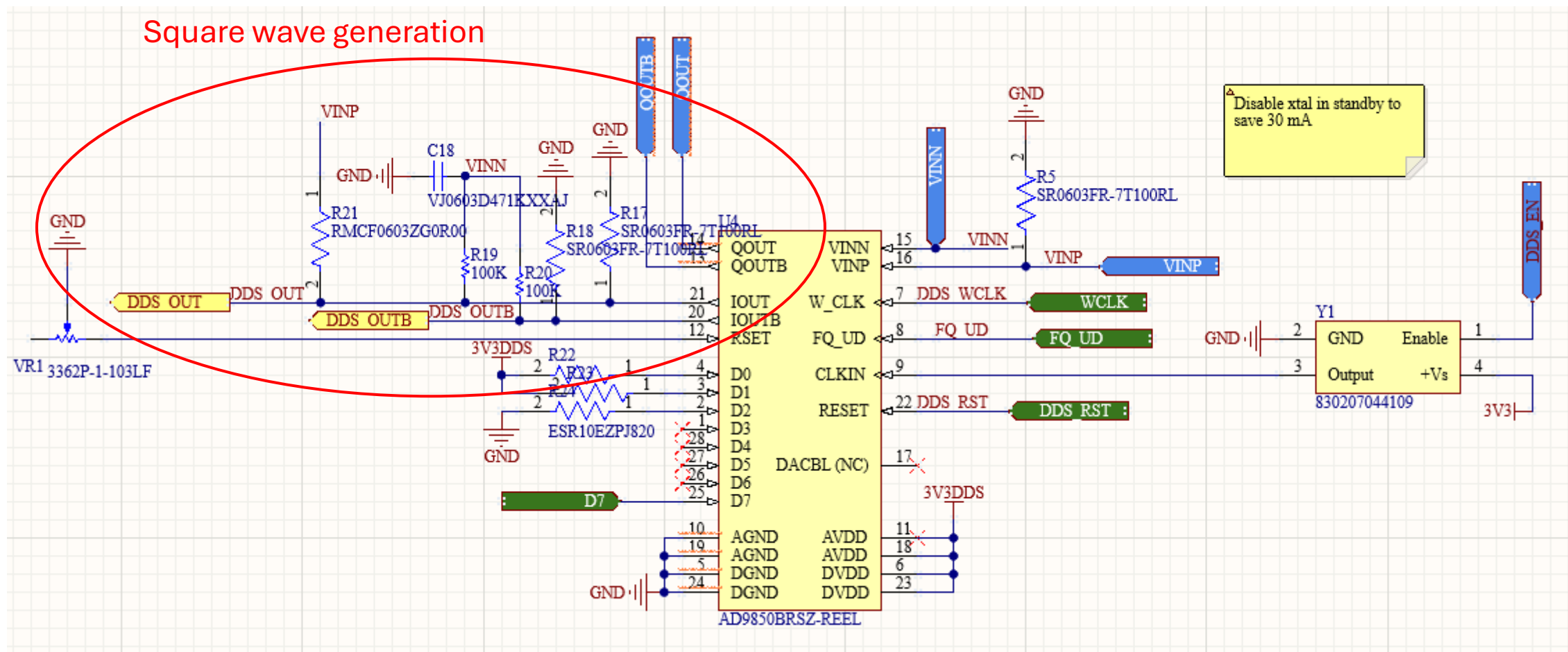
- The schematics were thoroughly reviewed by team members to find bugs.
- Changes were verified with datasheets and/or Ltspice.

Results:

- Schematic design was finished
- Some schematic blocks were updated compared to last week.
- PCB layout is 70% complete.

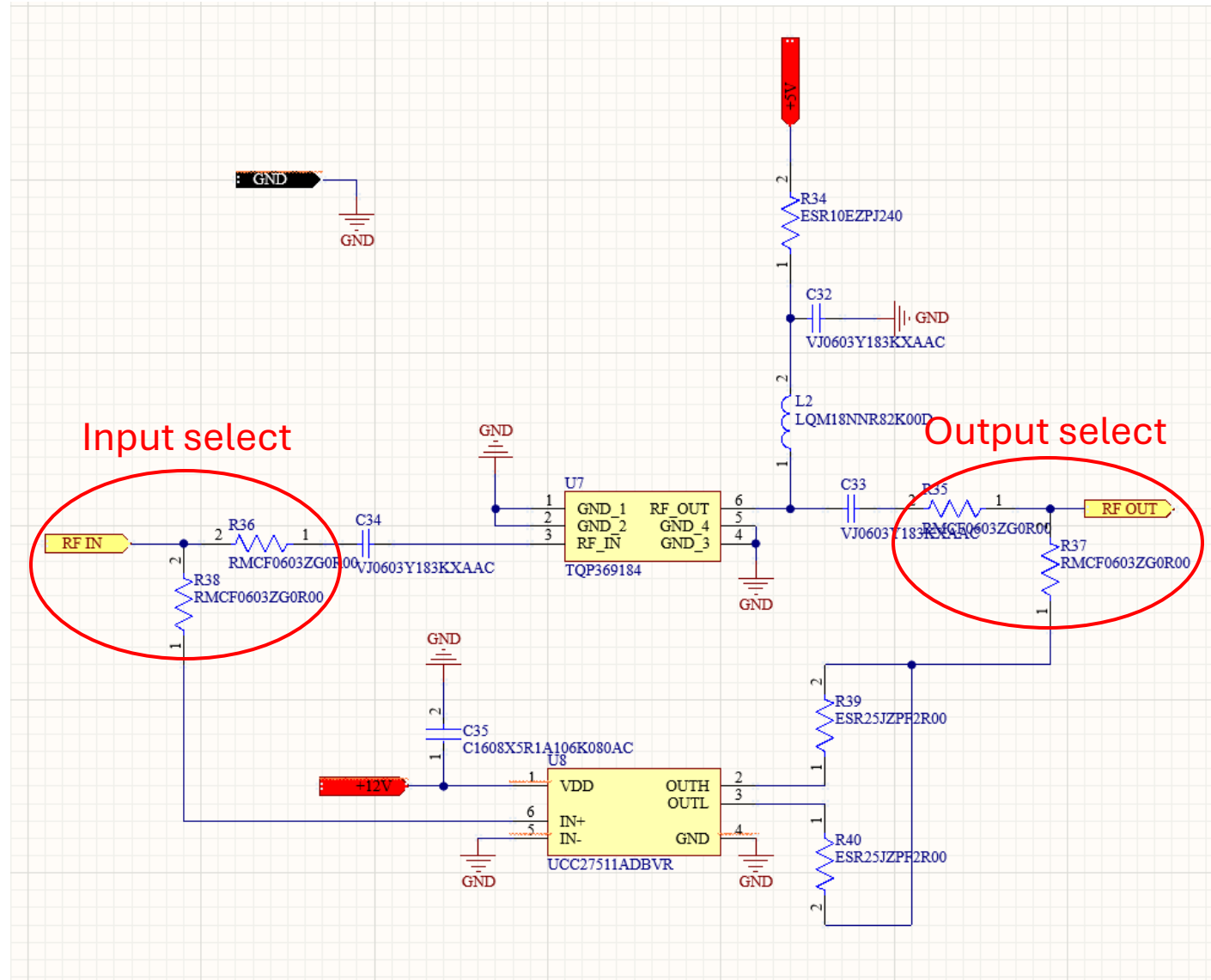
DDS design updates

- An option to use the comparator outputs and output a square wave was added for Class-E compatibility.
- 0 ohm resistors were added to provide choice of which outputs to use (sine vs. square).



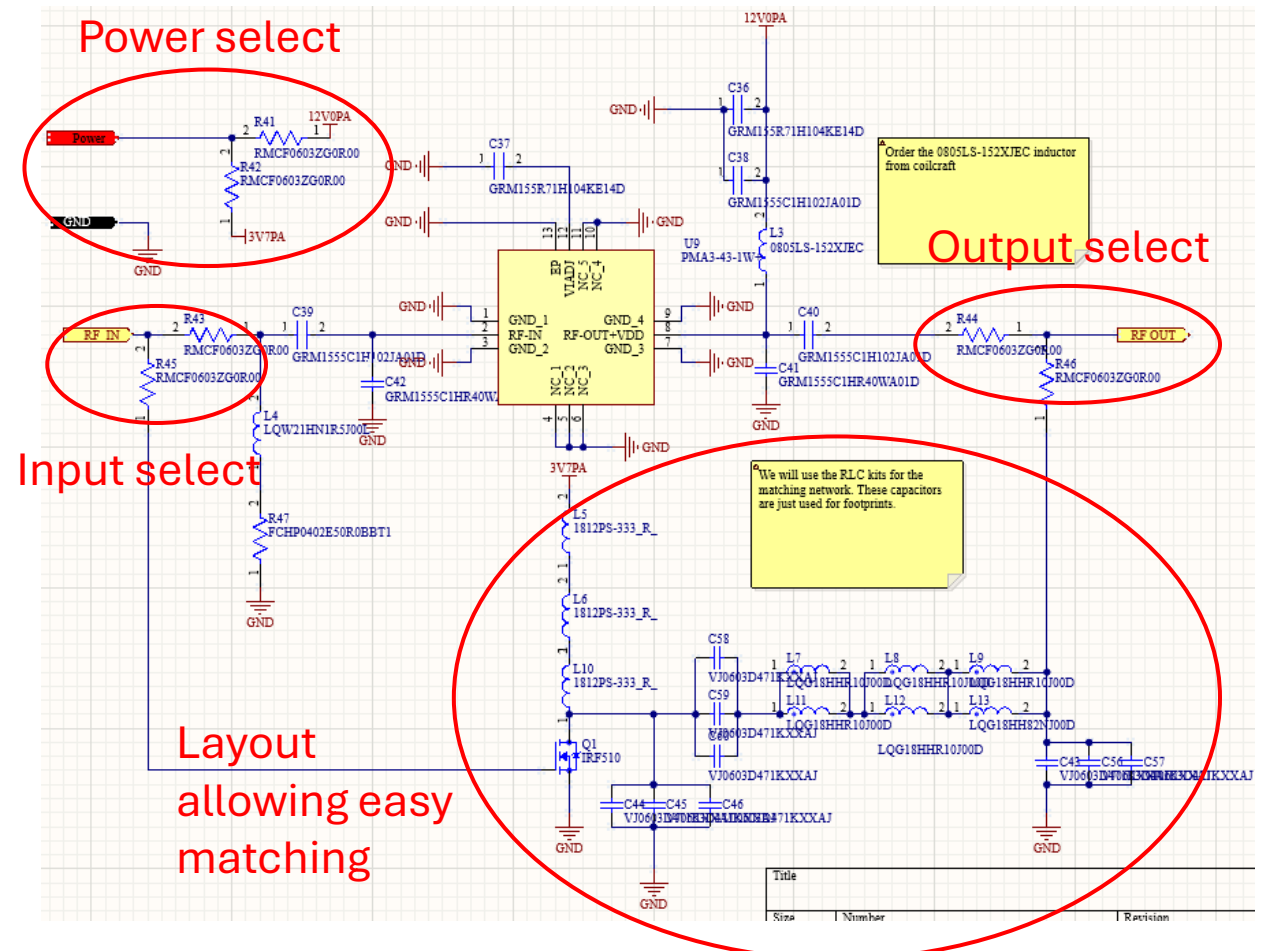
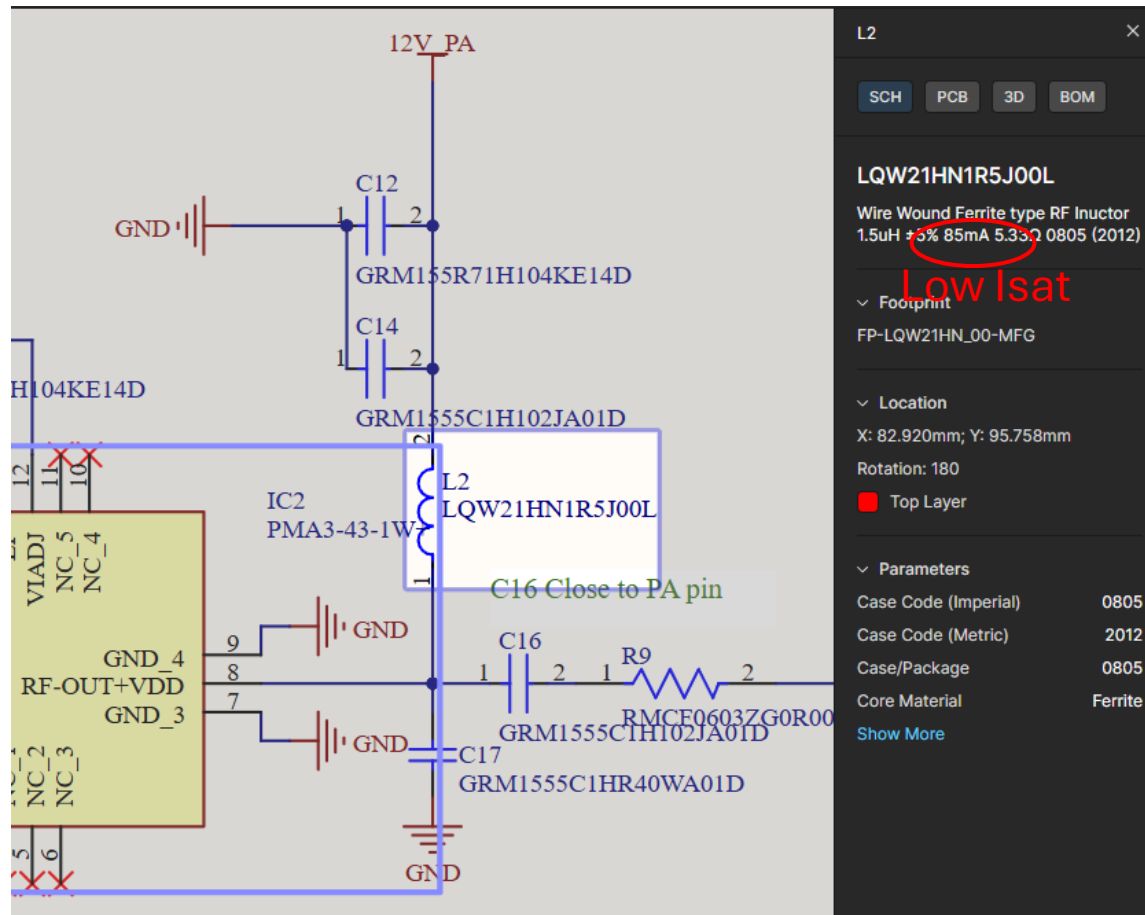
PA design updates: stage 1

- I placed the two 1stage PA designs in paralell with an option to use one or the other through 0 ohm resistors.
- These ICs will always be powered ON, because their static current consumption is low.



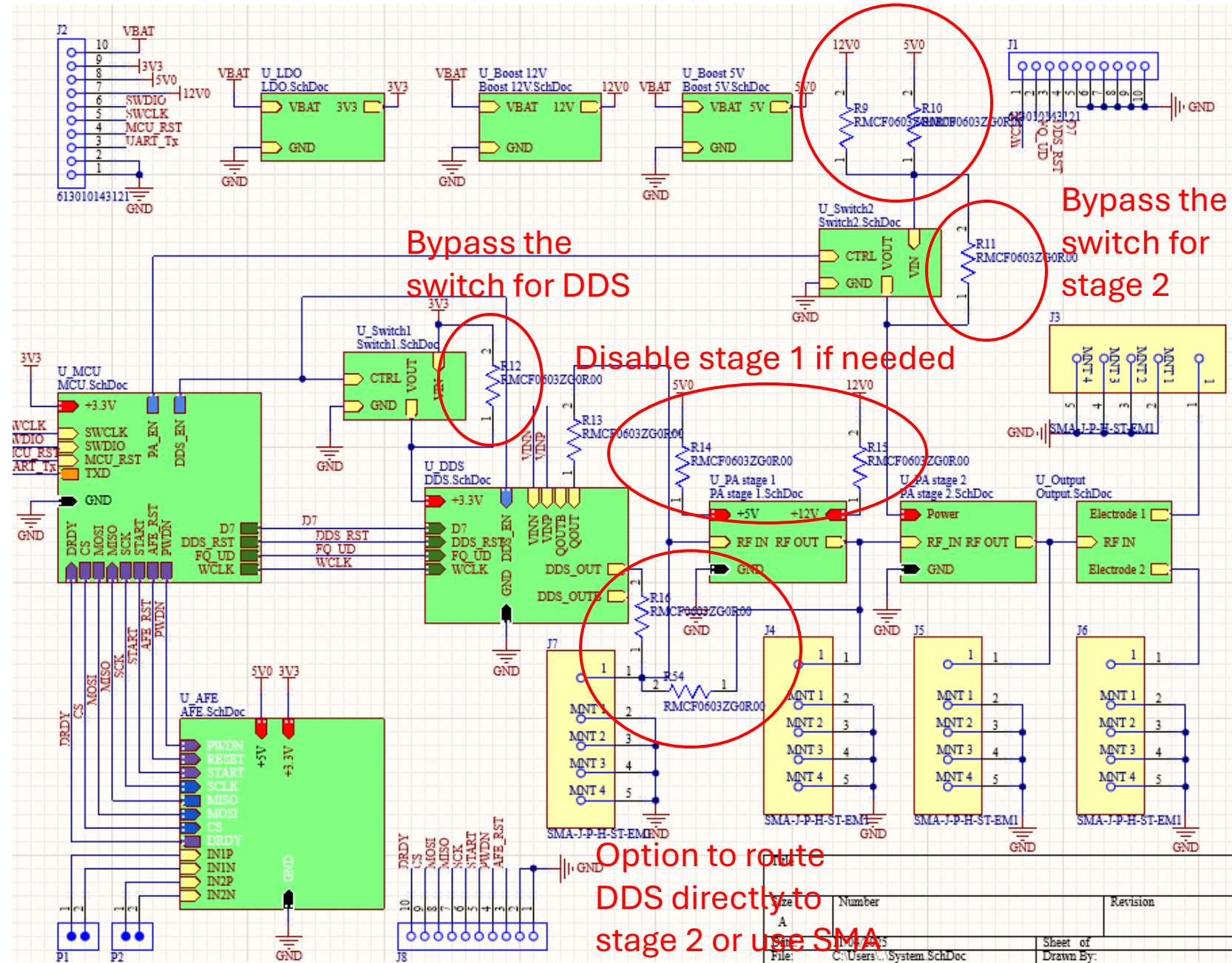
PA design updates: stage 2

- The output power inductor used previously wasn't the one recommended by the datasheet, which could explain lower performance. It saturates at only 85 mA. I replaced it with the recommended Coilcraft inductor rated for 400 mA.
- I designed the class-E stage 2 to allow for placing multiple capacitors / inductors in series or parallel to allow for tuning and matching.
- Since we have one switch but two different Vdd for stage 2, the power that's routed is selected with 0ohm resistors.



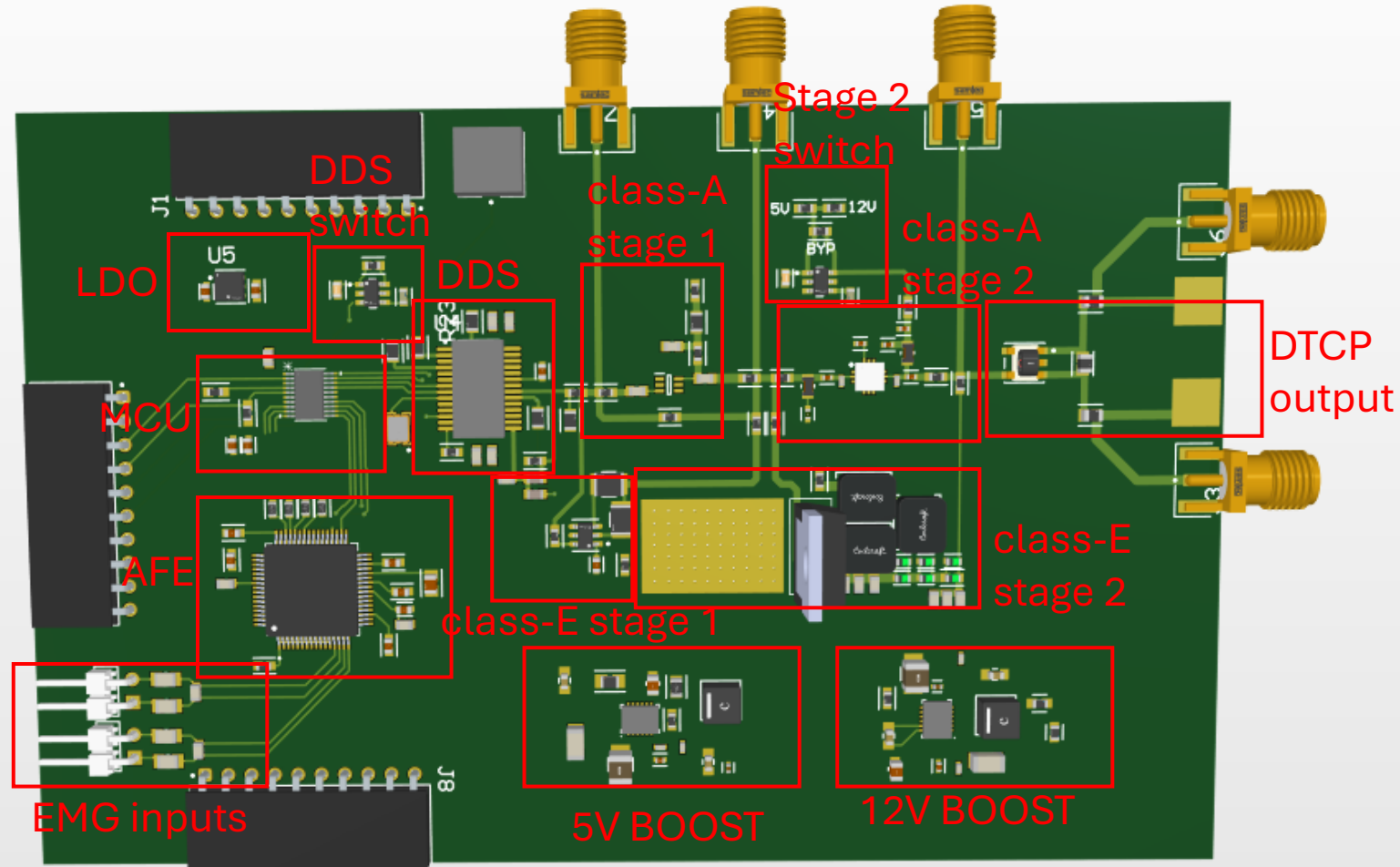
System schematic design updates

- A lot of consideration went into de-risking the design.
- Design blocks can be disconnected or bypassed without affecting other parts of the system.
- All important digital signals are available through debug headers
- All important analog signals are available through SMAs.



PCB layout updates

- Expected size: 80mm x 130 mm. Will be significantly reduced in the next PCB once we answer all the design using this one.



DTCP weeks 11/12 – Jeremiah Dados

Objective:

Finalize the PCB schematic and layout of the DTCP prototype. Place the fabrication order.

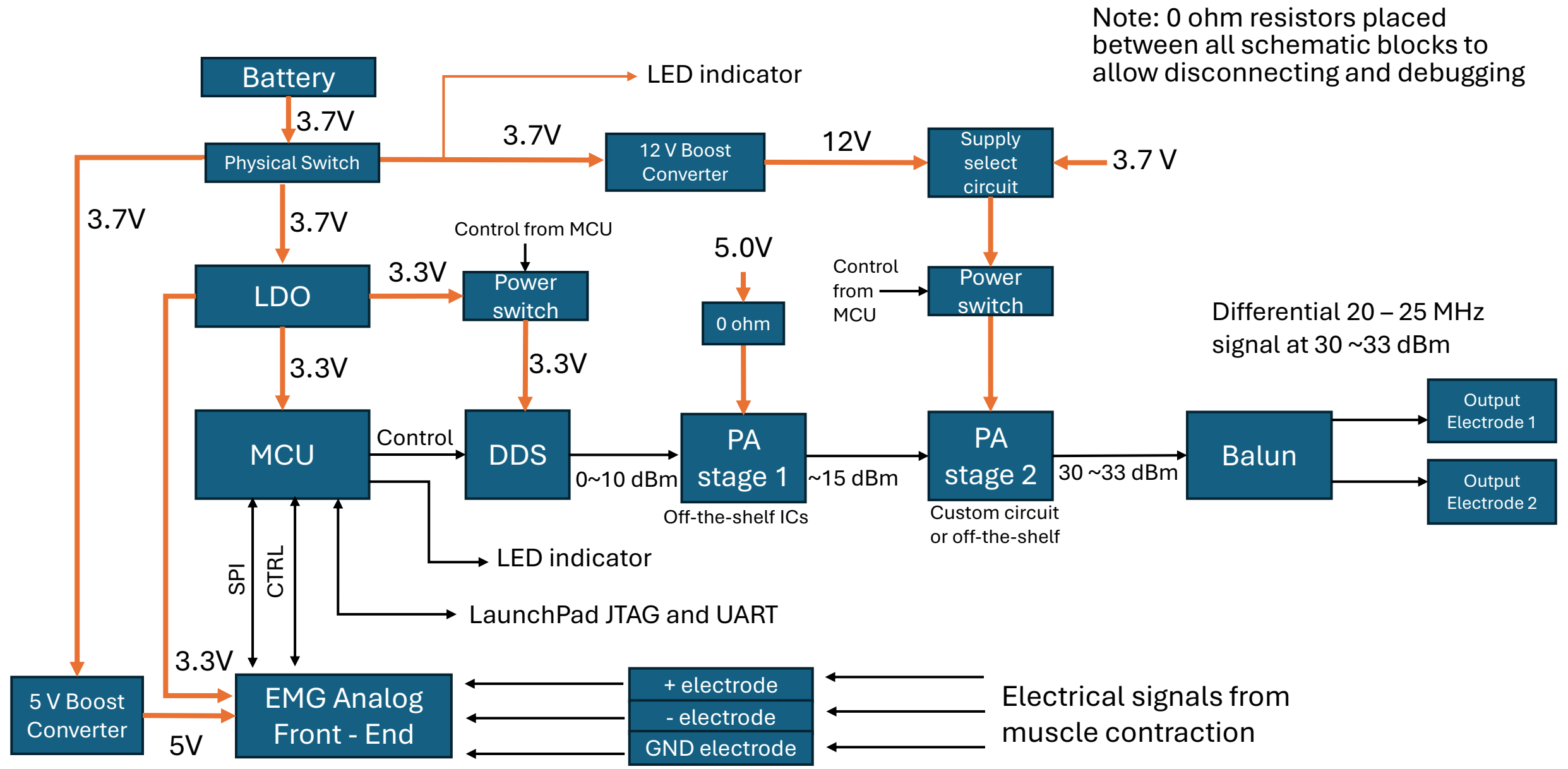
Methods:

- The schematics were thoroughly reviewed by team members to find bugs.
- Changes were verified with datasheets and/or Ltspice.
- Altium Designer used for layout.

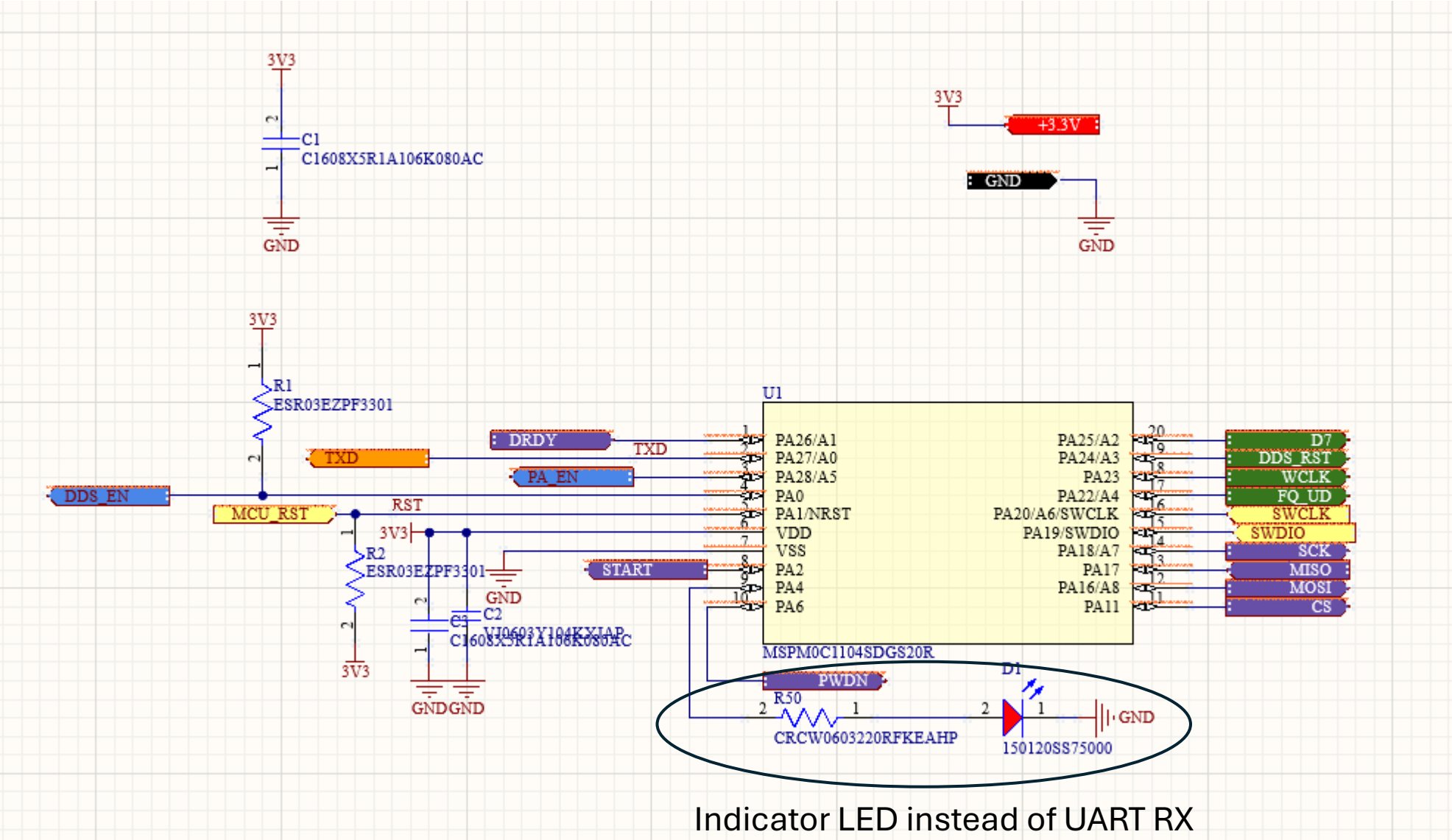
Results:

- Most schematic blocks as well as the system schematic were revised.
- PCB layout complete.
- Fabrication order placed

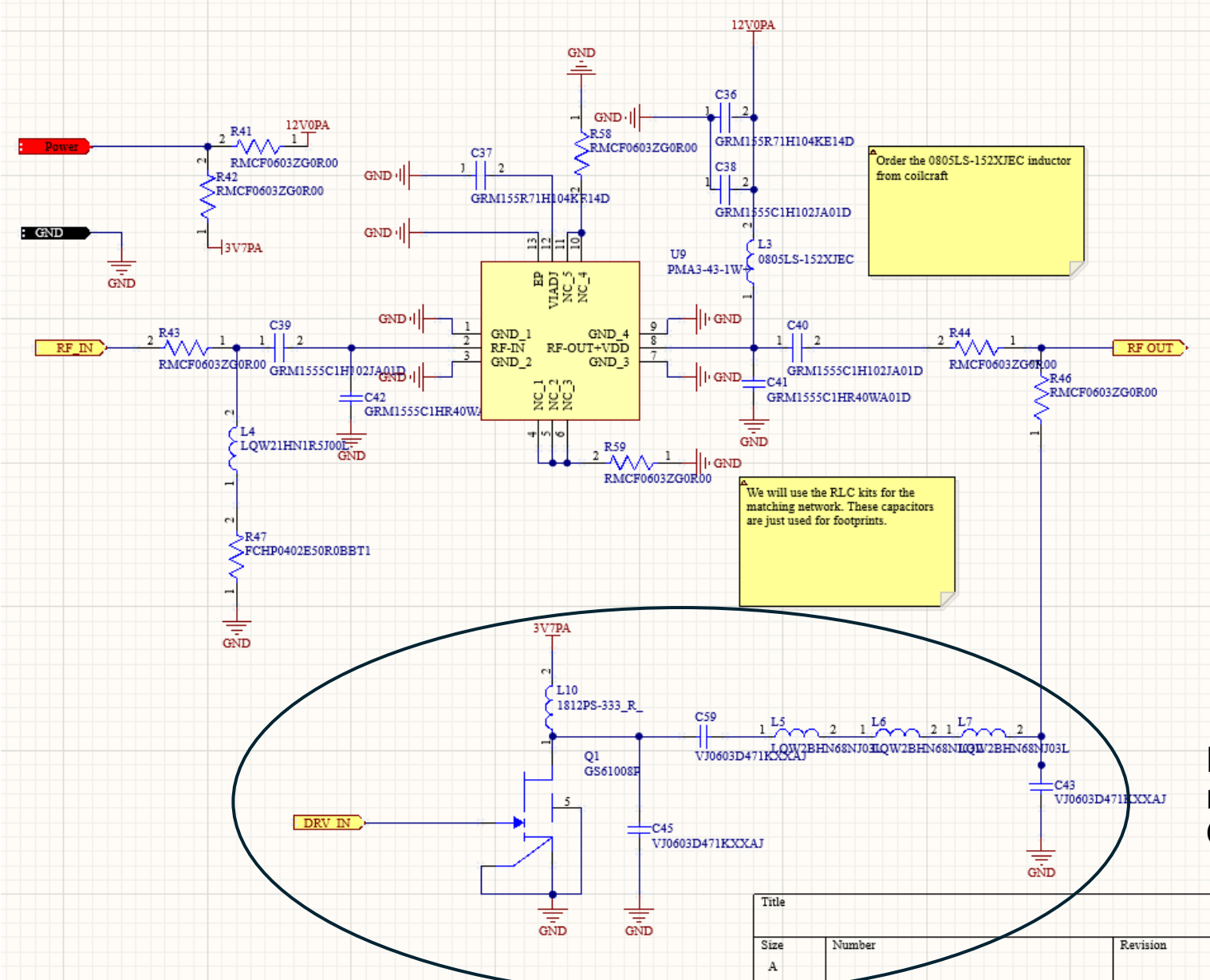
Fabricated system architecture



Updates to the MCU circuit

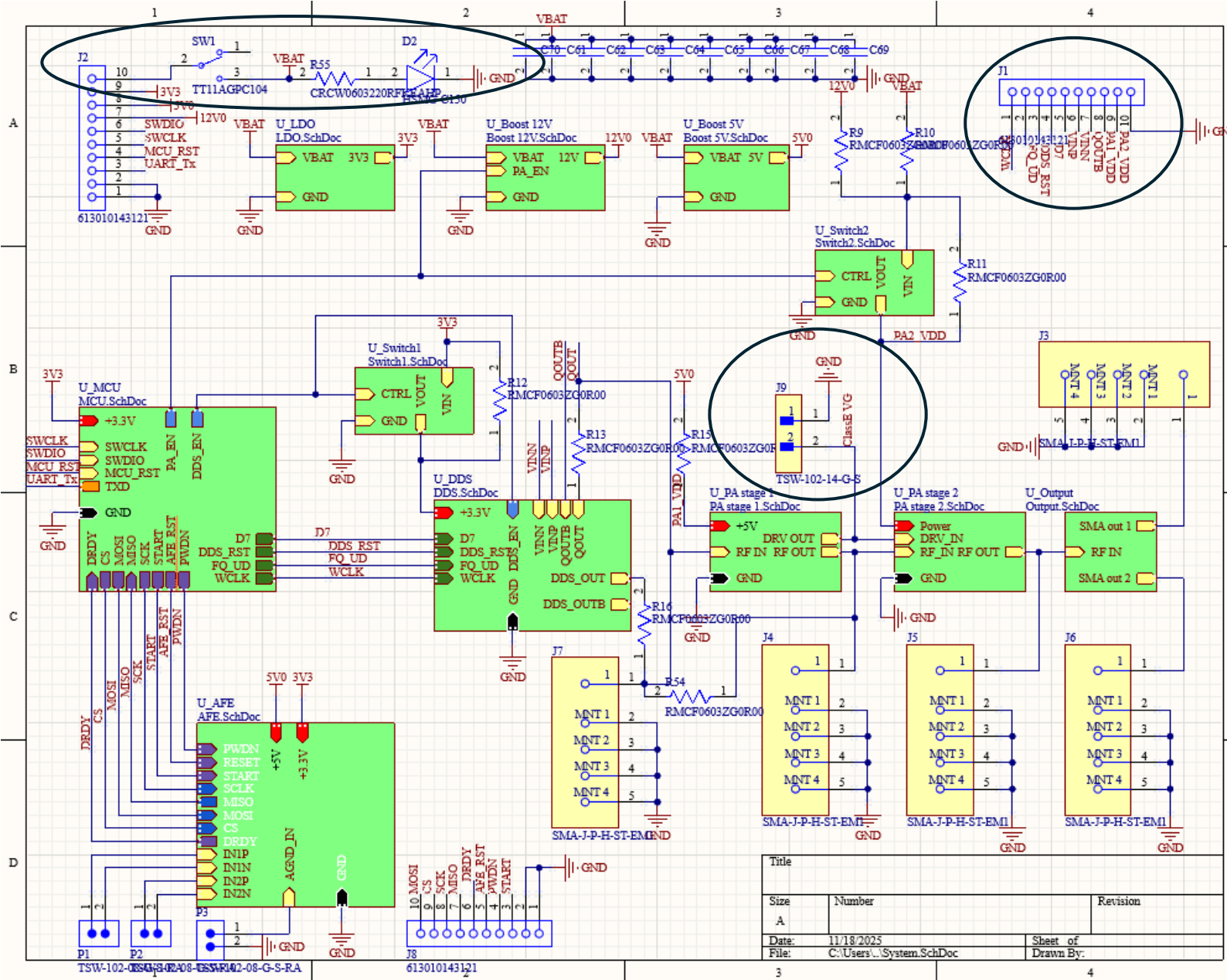


Updates to the PA circuit



Fabricated system-level schematic

Physical SPDT switch and power LED added



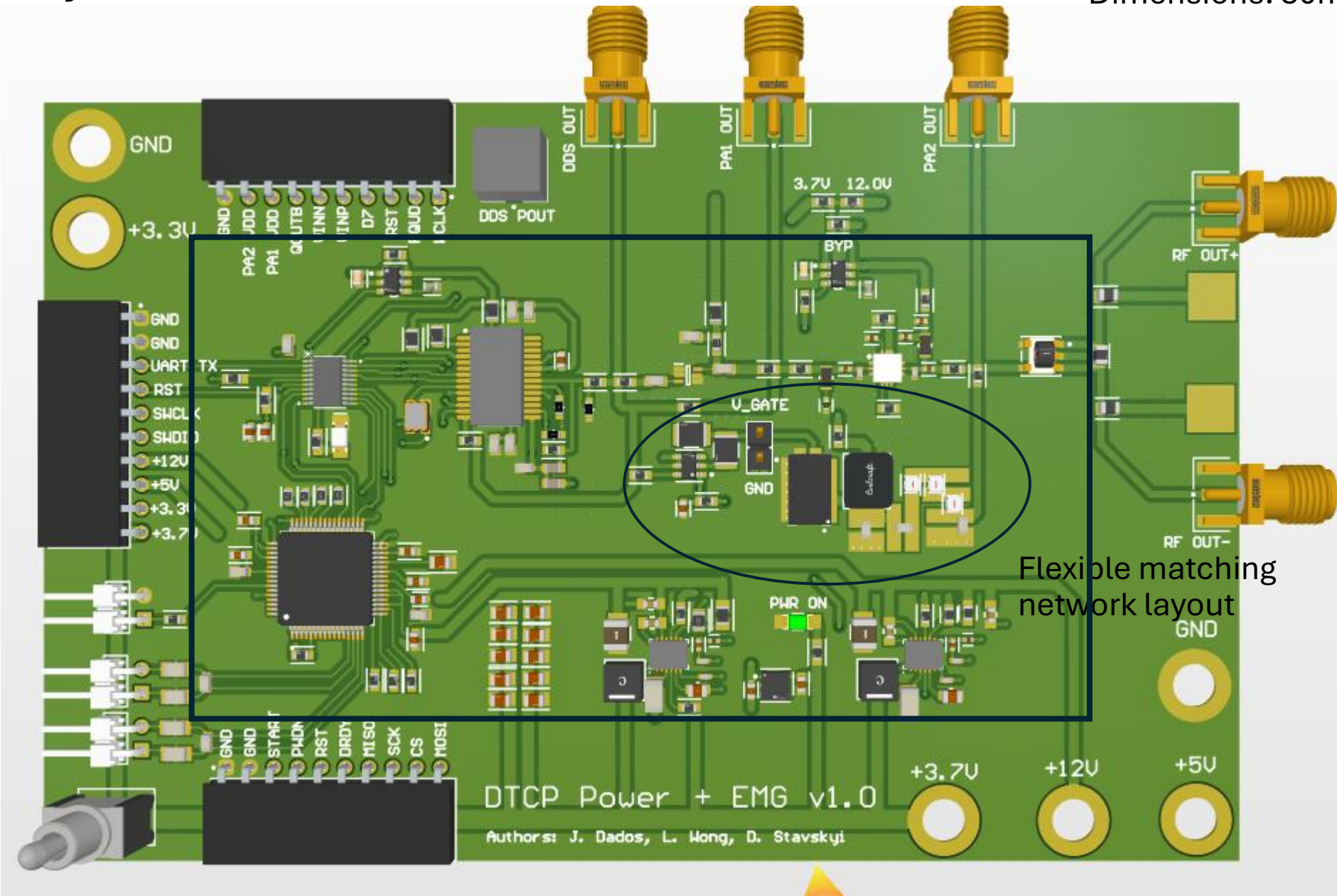
Debug connections added for PA drain voltages and comparator inputs and outputs

Direct access to class-E PA gate

Title		
Size	Number	Revision
A	11/18/2025	
Date:	11/18/2025	Sheet of
File:	C:\Users\...System.SchDoc	Drawn By:

Fabricated layout

Dimensions: 80mm x 130mm



DTCP weeks 13-15 – Jeremiah Dados

Objective:

Bring-up the new DTCP board.

Methods:

- Develop new firmware using code examples from Code Composer SDK.
- Assemble the PCBs in-lab with stencil and heatgun.
- Develop the testing plan and carry it out.

Results:

- Shown on the following slides.

Testing plan

Sub-system	MCU + control	DDS	PA (each stage)	Power circuits (LDO, buck)	AFE
Criteria for success	• Programmable from CCS • Current consumption <2 mA • Switch IC can open and short the circuits.	• Programs to correct output frequency • $P_{out} > 0$ dBm • $I_c < 90$ mA during TX and <2 mA in idle	• 2nd stage outputs >30 dBm • $I_c < 1.5$ A during TX and <10 mA in idle	• 12.0 V, 5.0 V, 3.3 V DC voltages generated • Negligible I_c of the power ICs	• SPI interface working (send and receive numbers) • Able to measure a 5 mVpp, 100Hz sine wave • $I_c < 50$ mA during meas. and <2 mA in idle
Validation	• Connect to the laptop and flash firmware. • Measure current using DC supply	• Measure F_{out} and P_{out} using SA • Measure current using DC supply	• Measure F_{out} and P_{out} using SA • Measure current using DC supply	• Measure DC voltages using oscilloscope • Measure current using DC supply	• SPI waveforms on oscpe in line with software data • Sine wave seen in debugger

Firmware development

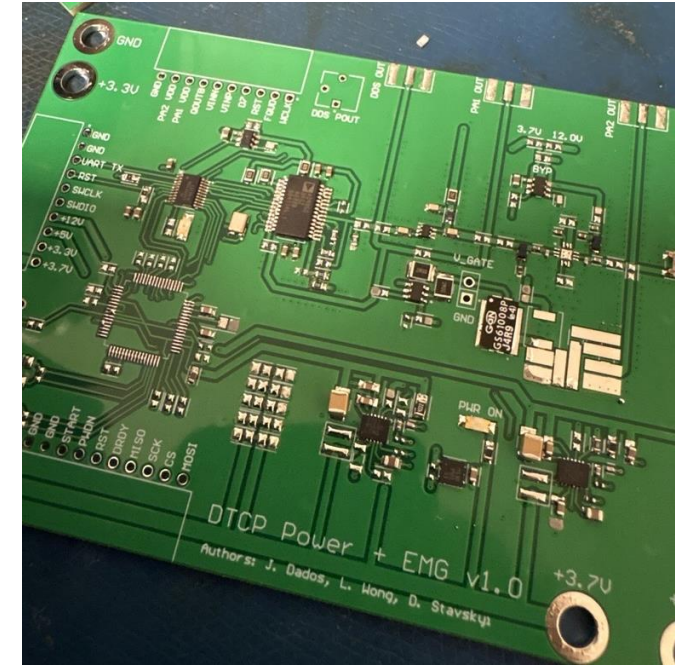
- Firmware is stored in a github repository
- I'm currently developing AFE access APIs

The screenshot shows the GitHub interface for the repository 'jdados / dtcp'. The top navigation bar includes links for Code, Issues, Pull requests, Actions, Projects, Wiki, Security, Insights, and Settings. A search bar is present on the right. Below the navigation bar, the repository path 'dtcp / firmware / DTCP_Firmware' is displayed. A commit history table is shown, listing files and their commit messages. The table has three columns: Name, Last commit message, and Last commit date.

Name	Last commit message	Last commit date
..		
.settings	Incorporated offset for register address during writing and tested th...	last week
.theia	Preliminary implementation of SPI (have not yet tested whether it out...	3 weeks ago
Debug	Commented out some things in the main	last week
src	Incorporated offset for register address during writing and tested th...	last week
targetConfigs	Empty CCS project to start the firmware development (using the latest...	last month
.ccsproject	Empty CCS project to start the firmware development (using the latest...	last month
.clangd	Empty CCS project to start the firmware development (using the latest...	last month
.cproject	Empty CCS project to start the firmware development (using the latest...	last month
.project	Empty CCS project to start the firmware development (using the latest...	last month
README.html	Empty CCS project to start the firmware development (using the latest...	last month
README.md	Empty CCS project to start the firmware development (using the latest...	last month
ads1299.h	Included the pins currently used on the board for SPI	last week
main.c	Commented out some things in the main	last week
main.syscfg	Cleanup the firmware repo	last week
uart.h	Can print a string with UART now	last week

Board assembly

- Because this is a very large circuit with many unknowns, I'm assembling the board and testing incrementally
- Order of testing: MCU, DDS, PA1, PA2, Power, AFE, Switching



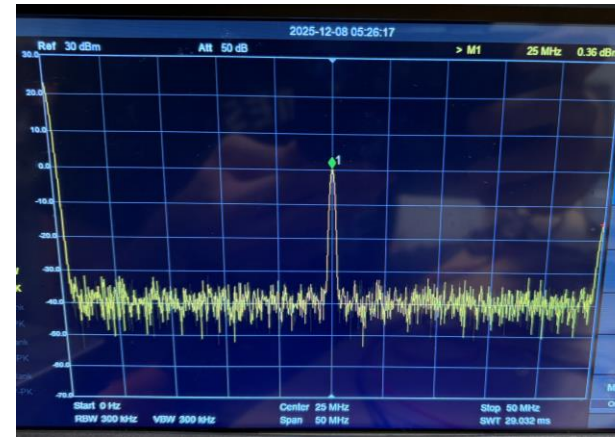
Objective:

Quantify the performance of the DDS circuit

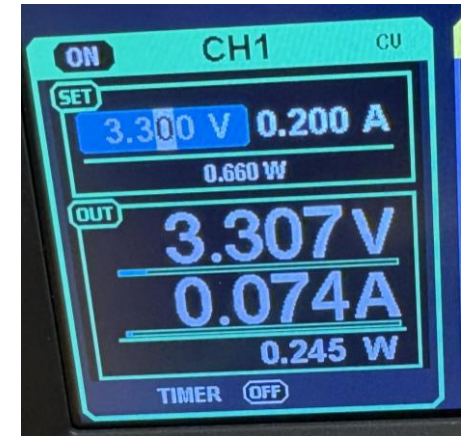
Methods:

- Measure current using DC supply.
- Measure the output power on SA.

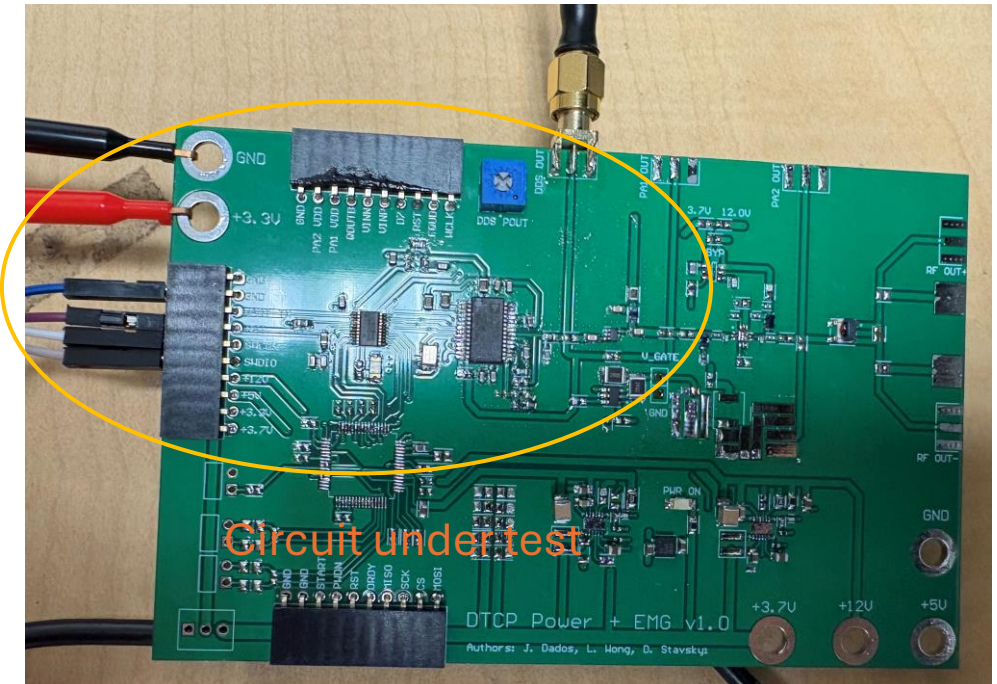
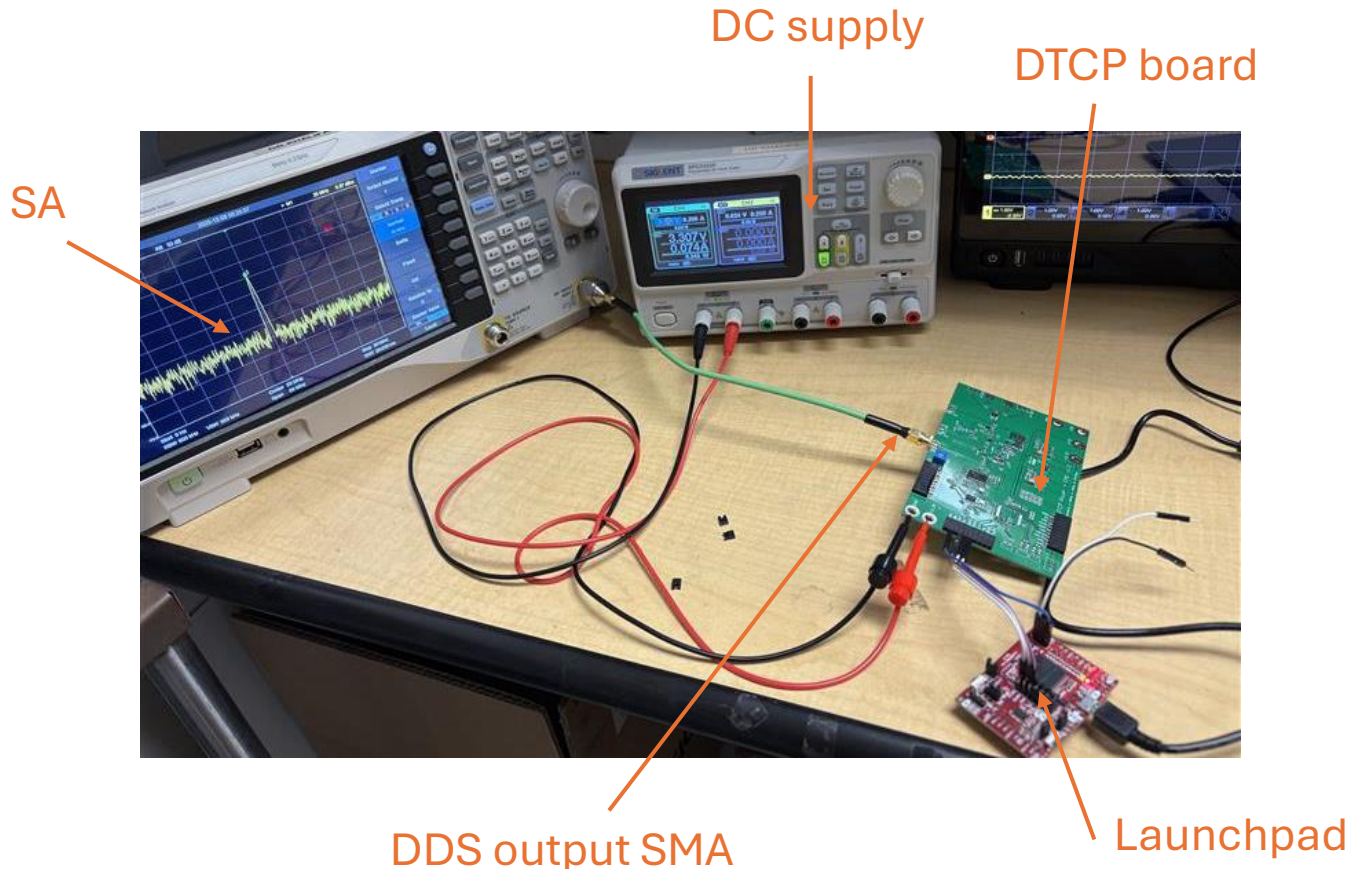
Results:



Pout = 0.36 dBm



Tx current



Results:



The DDS output signal is highly distorted, with very large harmonics stretching into the GHz range

The DDS chip is unstable, randomly dropping current consumption to 36 mA and having no output, with no change after swapping ICs or resoldering.

When the output gets connected to two PA stages, the chip reverts to the 36 mA state and the output disappears.

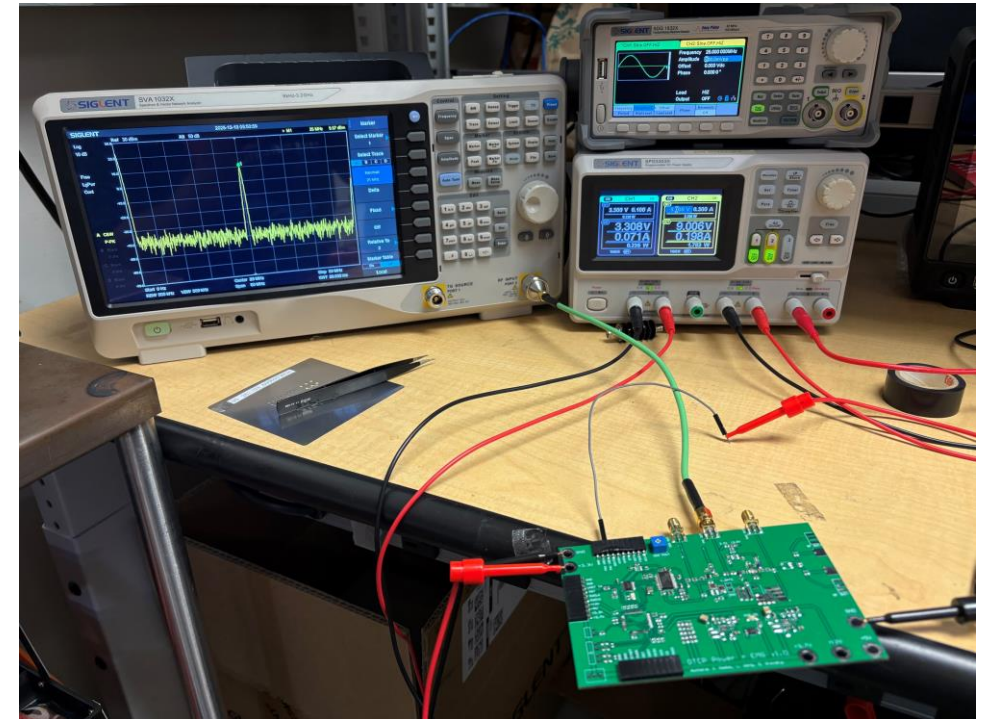
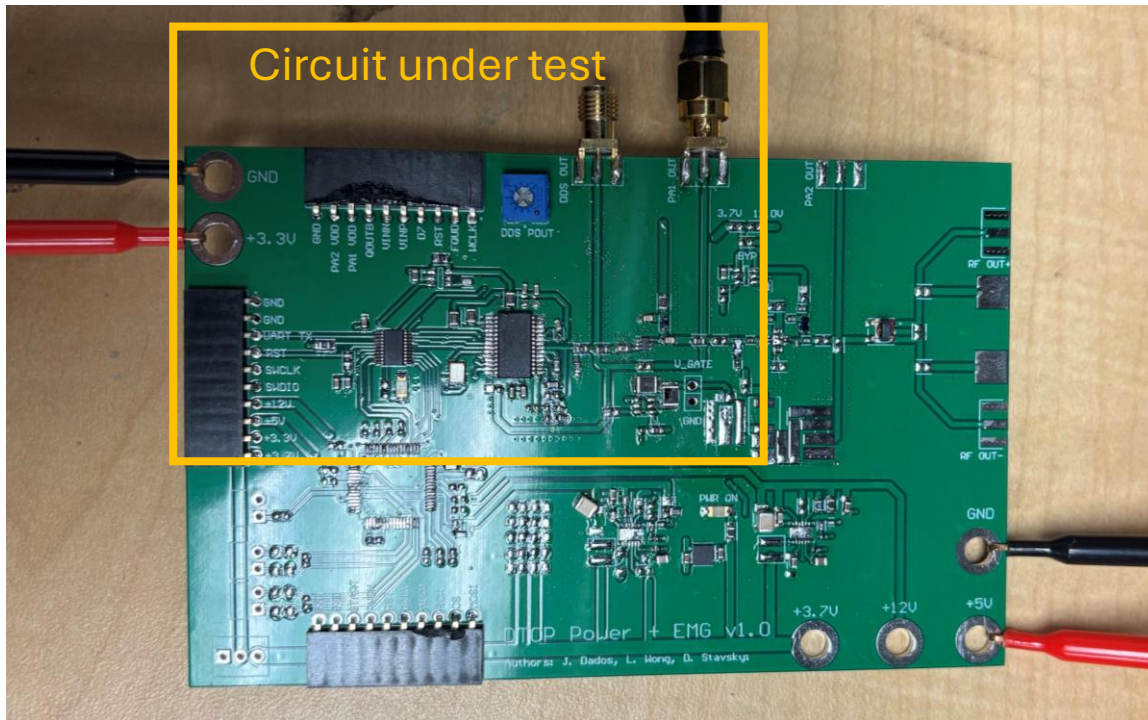
There is no output with no load resistor or with a 1k load resistor.

Objective:

Quantify the performance of the 1st stage class-A PA circuit

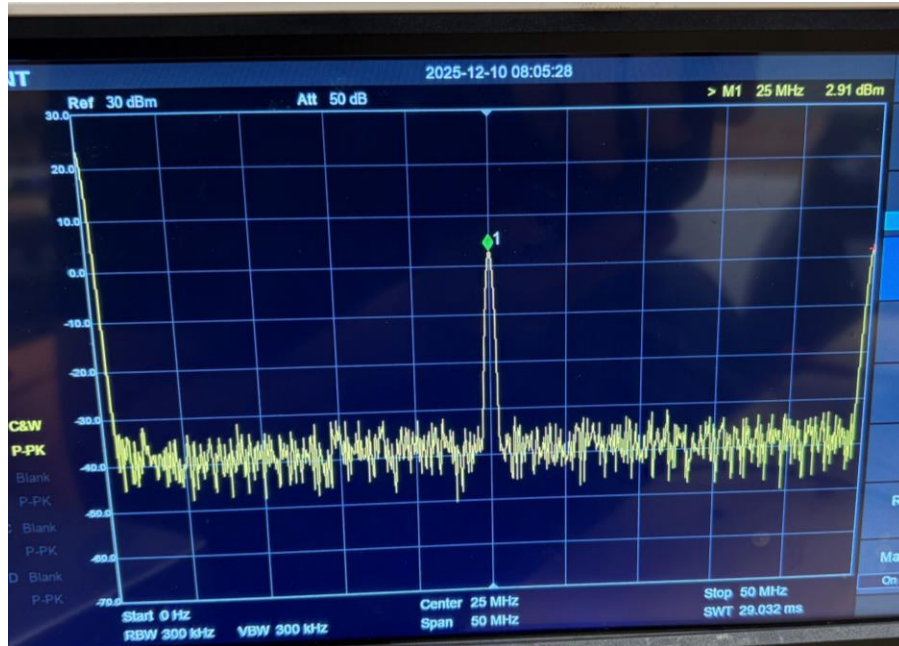
Methods:

- Measure current using DC supply.
- Measure the output power and frequency on SA.

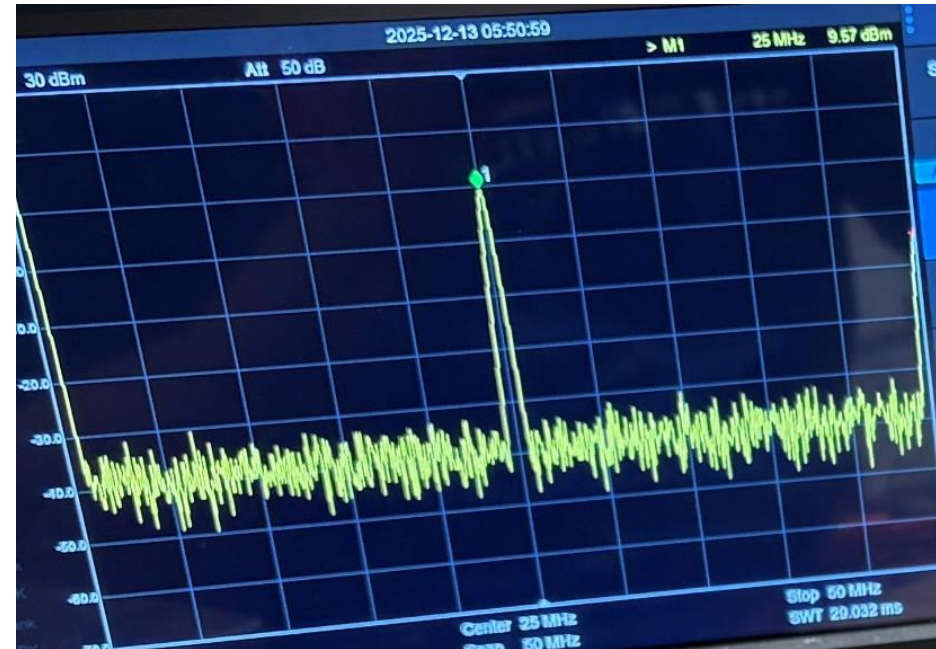


Setup

Results part 1:



At $V_{dd} = 5V$, the P_{out} is around 3 dBm



Highest P_{out} was at $V_{dd} = 9V$, reaching 9.5 dBm

Note- while sweeping V_{dd} up and down, two PA1 ICs became damaged and unusable



$I_q = 51 \text{ mA}$



DDS and PA1 Tx current at $V_{dd} = 5V$



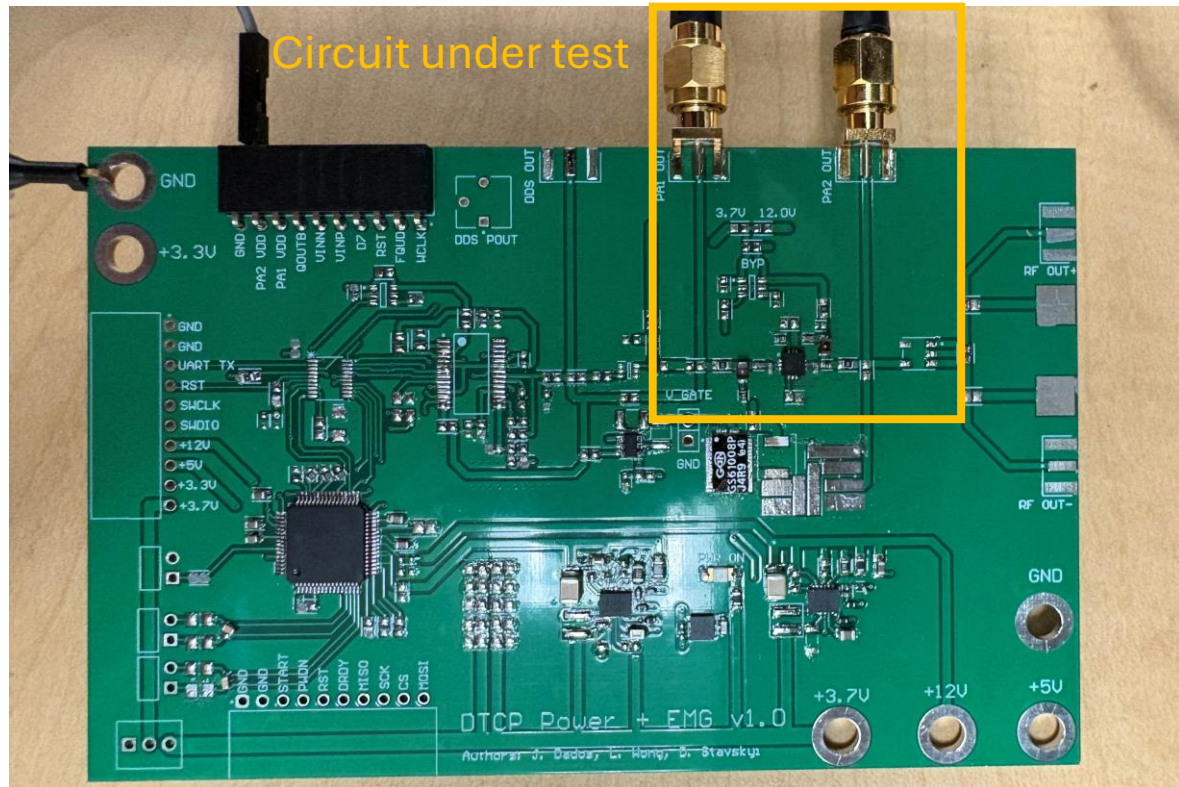
DDS and PA1 Tx current at $V_{dd} = 9V$

Objective:

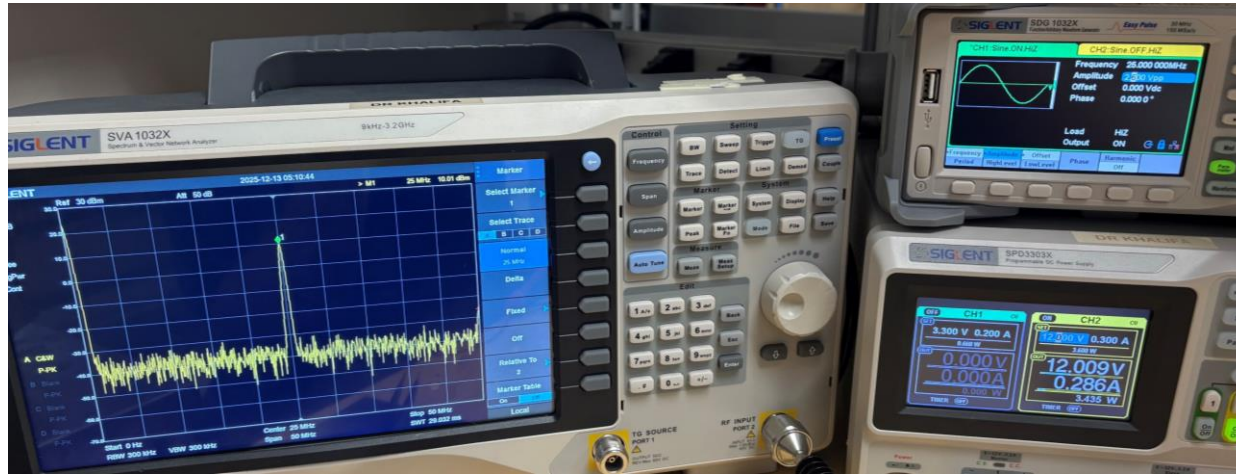
Quantify the performance of the 2st stage class-A PA circuit standalone

Methods:

- Measure current using DC supply.
- Measure the output power and frequency on SA.
- Input RF signals using the SDG1032 signal generator



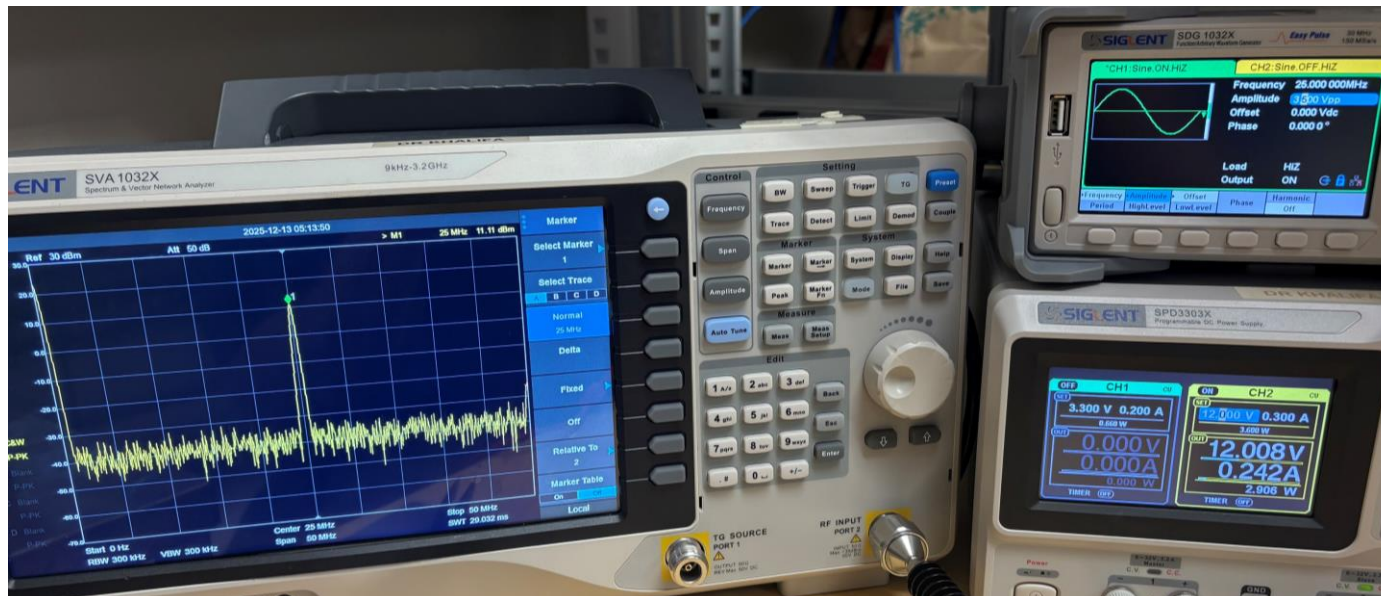
Results:



Pout = 30 dBm is achieved with Pin = 12 dBm supplied directly from a signal generator.

The PA is very difficult to solder. In the process of testing, one of the PA chips became damaged, creating a short on the PCB I haven't been able to get rid of despite hours of trying.

Because of that, I had to assemble a new PA2 circuit on a separate PCB and collected all the data with it, inputting signals through the input side SMA.



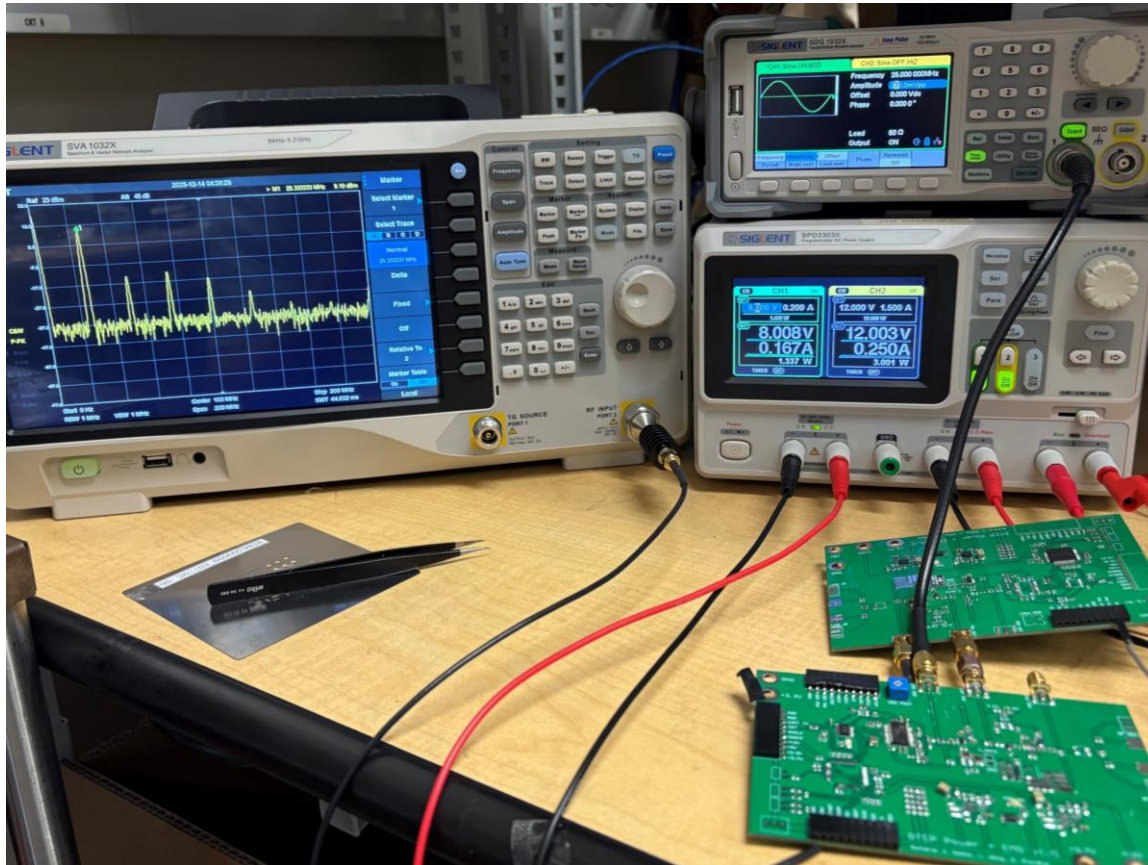
Pout = 31.1 dBm is achieved with Pin = 15 dBm supplied directly from a signal generator.

Objective:

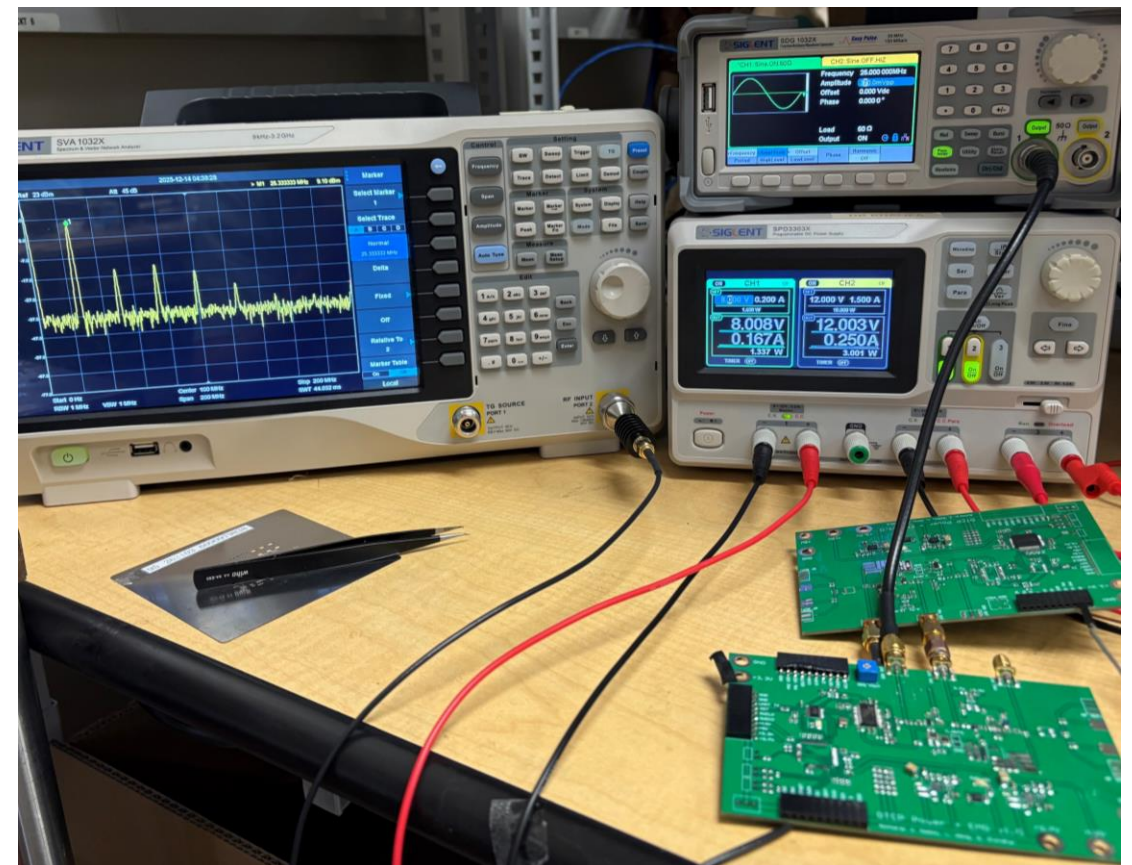
Quantify the performance of the 2st stage class-A PA circuit in conjunction with the 1st stage class-A PA

Methods: Same as before

Results :



$P_{out} = 25 \text{ dBm}$ was achieved with
 $P_{in} = -5 \text{ dBm}$ and PA1 Vdd = 5V



$P_{out} = 29.1 \text{ dBm}$ was achieved with
 $P_{in} = -5 \text{ dBm}$ and PA1 Vdd = 8V

Next steps:

- Find a replacement for the DDS
- Test the class-E power amplifier circuits
- Test the AFE circuit
- Test the DDS with Pout set to less than max

Spring 2026

Objective:

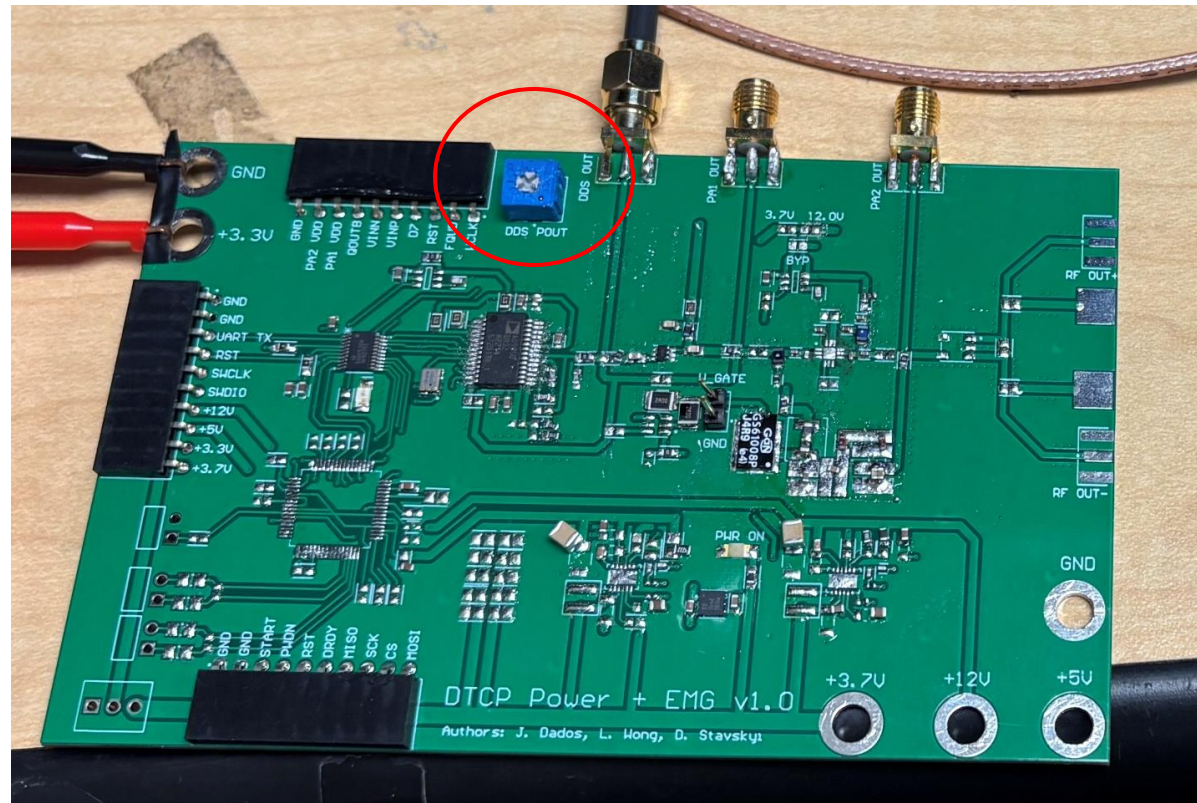
Determine the cause of the DDS circuit not working reliably / not producing output.

Methods:

Measure digital waveforms on oscilloscope, measure current consumption on the power supply, visually inspect the soldering on the PCB

Results :

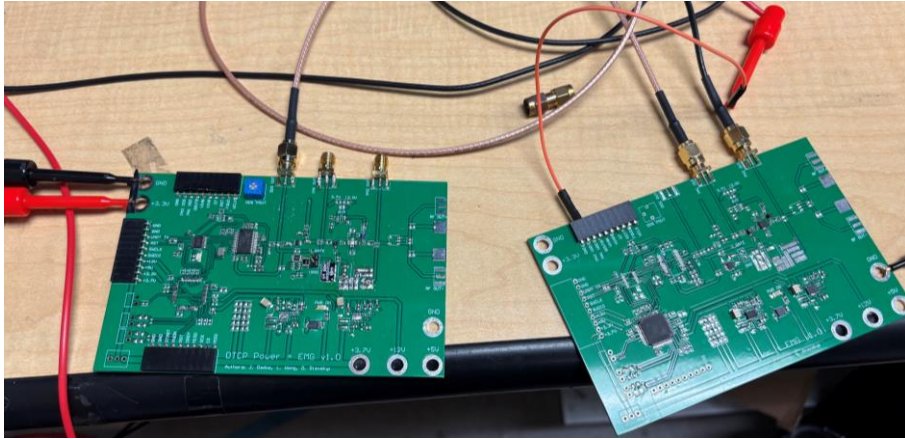
Mechanical failure of the potentiometer was found to be the cause of the problem. The component was likely damaged by heat from resoldering.



Objective:

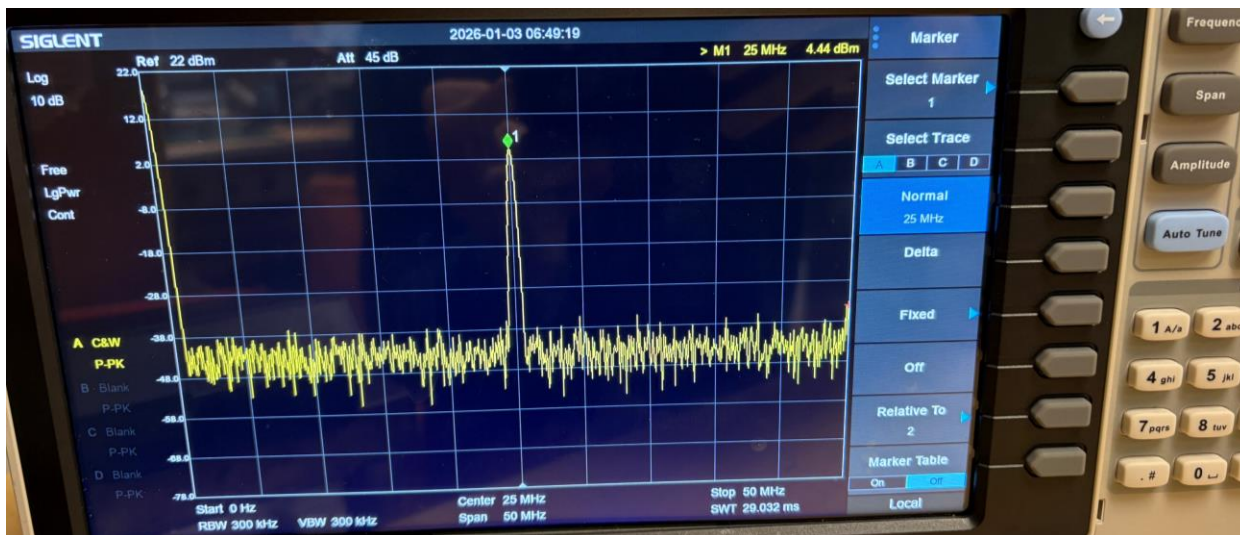
Determine the maximum output power with DDS connected directly to the 2nd stage PA.

Methods:



DDS on one PCB, 2nd stage PA on another. Both powered from DC power supply.

Results:

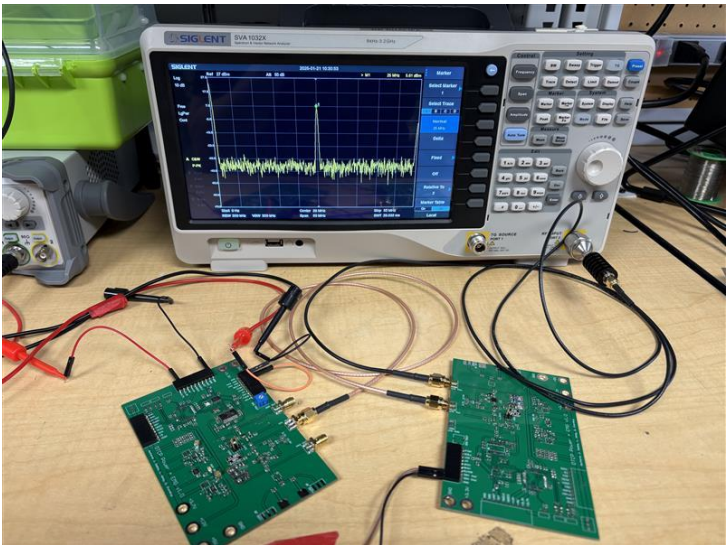


24.44 dBm achieved (20 dB attenuator used)

Objective:

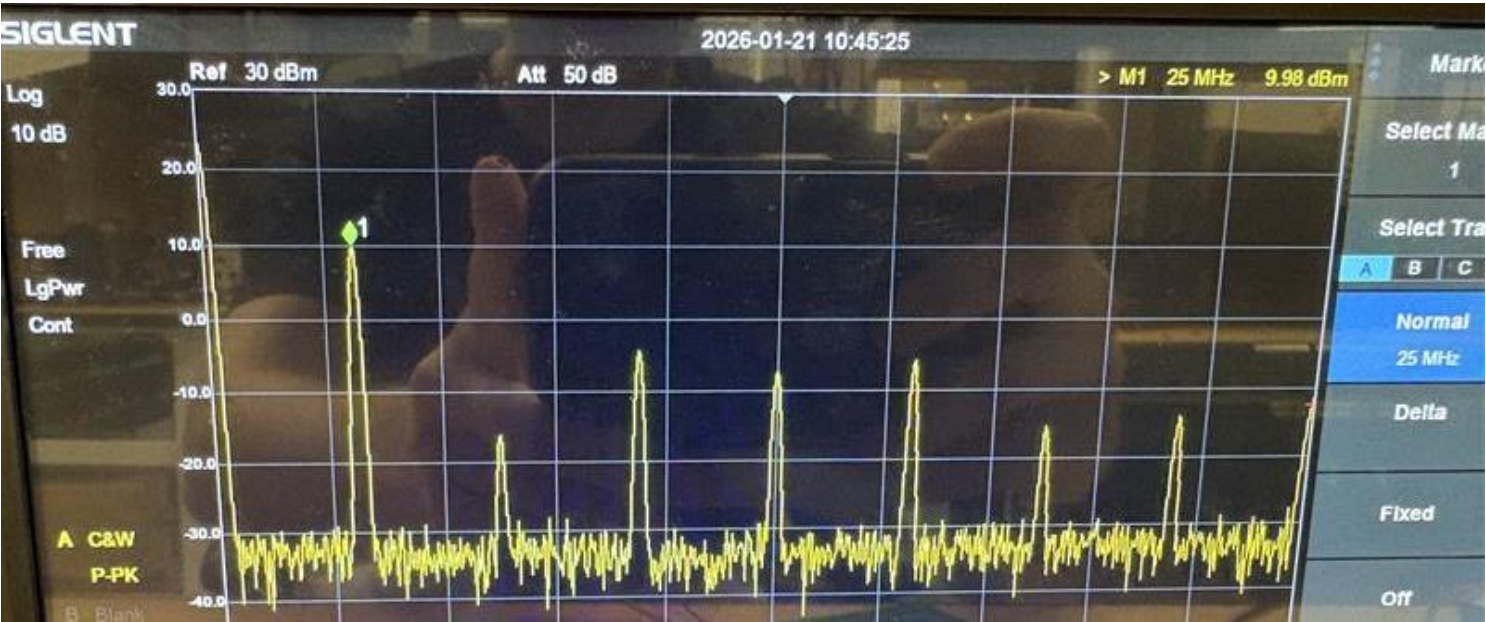
Determine the maximum output power of the full class-A signal chain (DDS + PA1+ PA2).

Methods:



DDS + PA1 on one PCB, 2nd stage PA on another. Both powered from DC power supply.

Results:



30 dBm achieved with changes to PA supply voltages

P_out	PA1_VDD	PA2_VDD
27.7 dBm	5.0 V	12.0 V
29.5 dBm	7.0 V	12.0 V
30.0 dBm	7.0 V	12.5 V

Summary of P_out vs supply voltage

Updated project status and timeline

1. RF transmitter circuit status

- The class-A PA circuit is fully functional, but requires modifications of the DC supply voltages.
- The custom class-E circuit was assembled, but is extremely inefficient in the current configurations

2. Next steps

- Determine how the on-board supply voltages can be adjusted to allow for 30dBm output with the class-A design.
- Experiment with different matching network components in the class-E circuit until >30 dBm is achieved at 25 MHz.
- Place the power switches on the board and validate if they can switch PA stages on and off to limit the static current consumption.

Updated project status and timeline

1. EMG recording circuit status

- We assembled the AFE circuit on a separate board and a Launchpad is being used to program it.
- As of right now, we established and verified SPI communication with the chip, but haven't obtained voltage readings yet.

2. Next steps

- Measure a 10 mVpp, 100 Hz sine wave using a signal generator connected to the AFE inputs.

Spring 2026 timeline

Second Prototype

The second prototype addresses issues found in the first design and focuses on improving compactness, flexibility, and performance.

Task	Finish By	Leader	Supporters	Comments	Status
Revisions to the circuit design	Mid-February 2026	Luis Wong, Dmytro Stavskyi	Jeremiah Dados		✗
Design and order the PCB	End of February 2026	Jeremiah Dados	Luis Wong, Dmytro Stavskyi		✗
Place the fabrication order	End of February 2026	Jeremiah Dados	Han Wu		✗
Receive the board and assemble it	Mid-March 2026	Luis Wong, Dmytro Stavskyi	Jeremiah Dados		✗
Validate board outputs on lab equipment	End of March 2026	Jeremiah Dados	Luis Wong, Dmytro Stavskyi		✗
Validate board functionality in animal experiments	End of April 2026	Han Wu	Jeremiah Dados, Luis Wong, Dmytro Stavskyi		✗

Target specifications

Performance metric	Target value	Measured value as of 1/28/2026	Explanation
Output frequency	25 MHz	25 MHz	Measured with an RF spectrum analyzer.
Output power	≥30 dBm differential	30.7 dBm	Measured with an RF spectrum analyzer. 30 dBm differential = 27 dBm per electrode.
Static current	<10 mA	360 mA (without power switches)	Current consumed when the RF signal is not being transmitted. Needed for 12 hr battery life with a 140 mAh battery.
Tx current	<500 mA	430 mA	Current consumed when the RF signal is being transmitted.
EMG noise floor	<0.5 mV	N/A	EMG spikes reach up to +5 mV, which would give us SNR of 10 dB.
EMG sampling rate	≥1000 samples/s	N/A	Enough to measure signals up to 500 Hz.
Board dimensions	50 mm x 70 mm		About the same size as ssEEG

Objective:

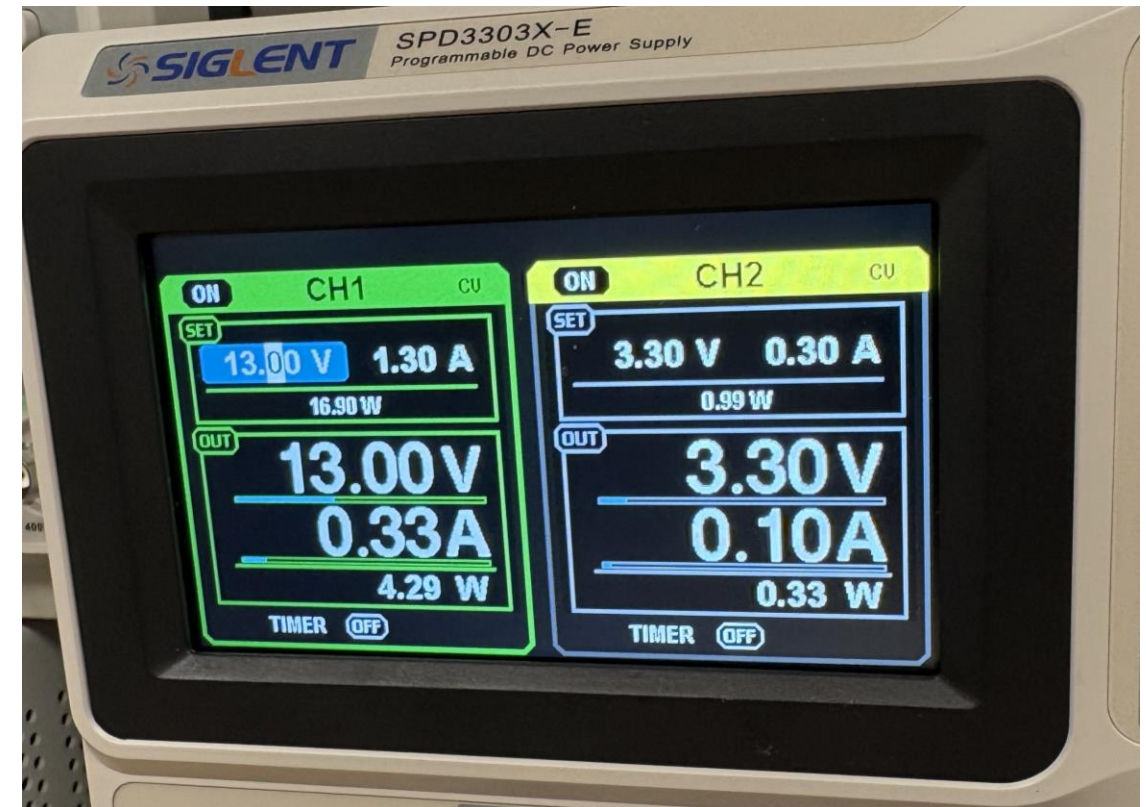
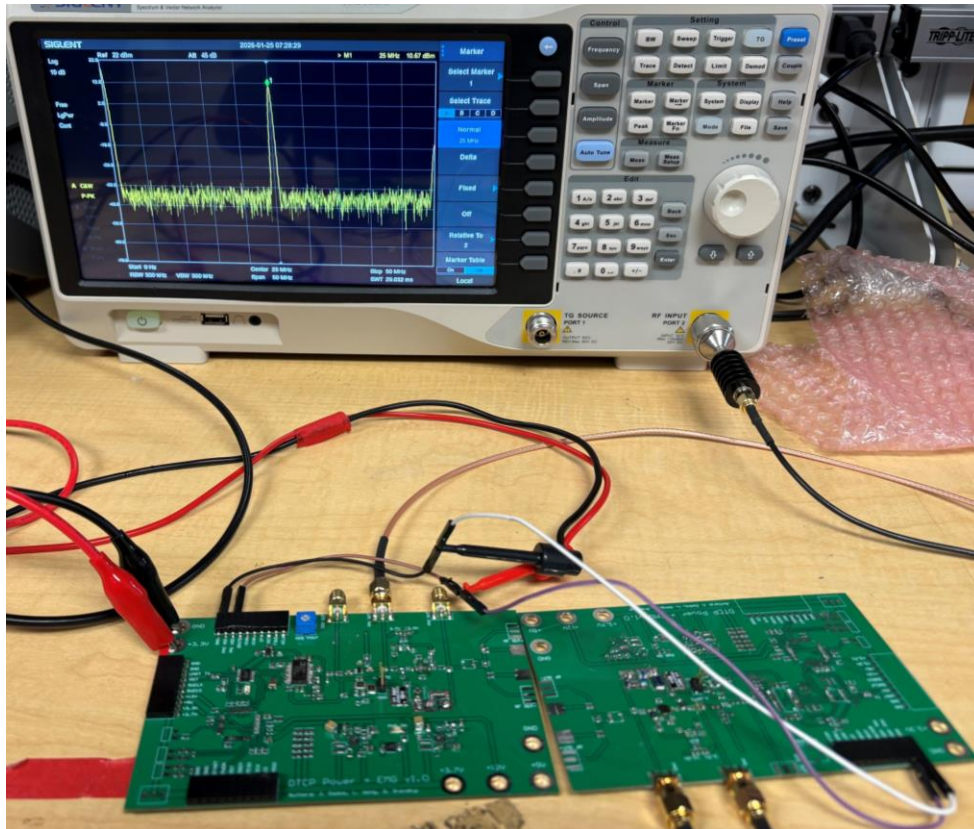
Determine the class-A PA1 drain resistor value needed to use one power supply for both PA1 and PA2.

Methods:

Start with $R = 100\ \Omega$ and decrease resistance until sufficient output power is achieved at $V_{dd} = 12 \sim 13\ \text{V}$.

Results:

$R=51\ \Omega$ yielded best performance. $V_{dd}=13\text{V}$ for both PA gives output power of 30.7 dBm delivered to the balun.



Objective:

Determine the insertion loss of the balun.

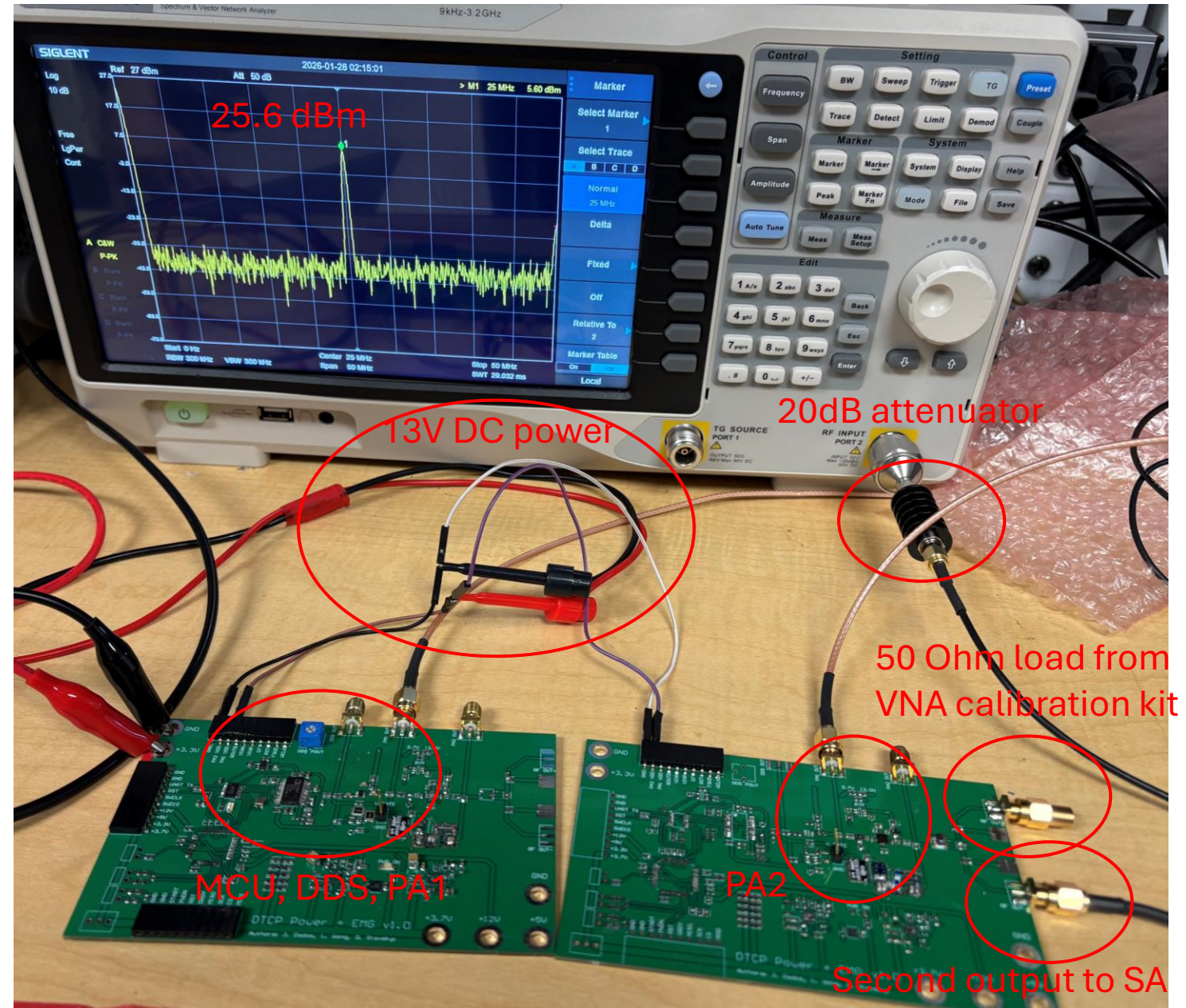
Methods:

Connect a 50 ohm load to one differential output, measure the power output of the other one.

Results:

Each differential signal is 25.6 dBm, meaning the insertion loss is 2.1 dB.

Theoretical value with no insertion loss:
 $30.7 \text{ dBm} - 3 \text{ dBm} = 27.7 \text{ dBm}$



Objective:

Determine whether the power switches operate as expected

Methods:

Update the firmware to assert/de-assert the connections to the control pins of the power switches. Observe power consumption as power to the DDS is turned on and off.

Results:

Power consumption of the 3.3V supply drops to <1 mA after the switch control is de-asserted.

Next steps

- Assemble a complete RF transmitter PCB.